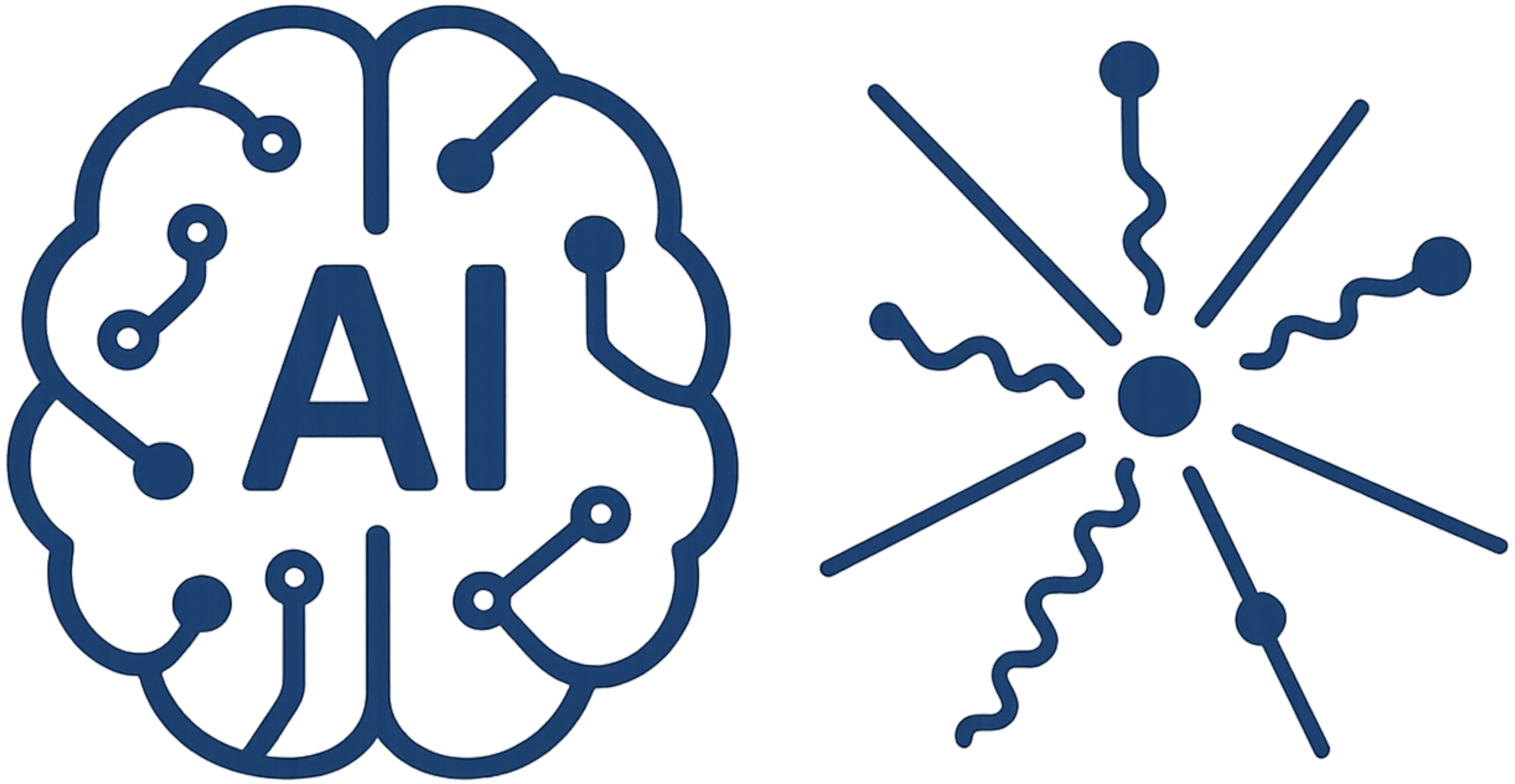
Introduction to

AI-Driven HEP

S. A. Fard - School of Physics (IPM)



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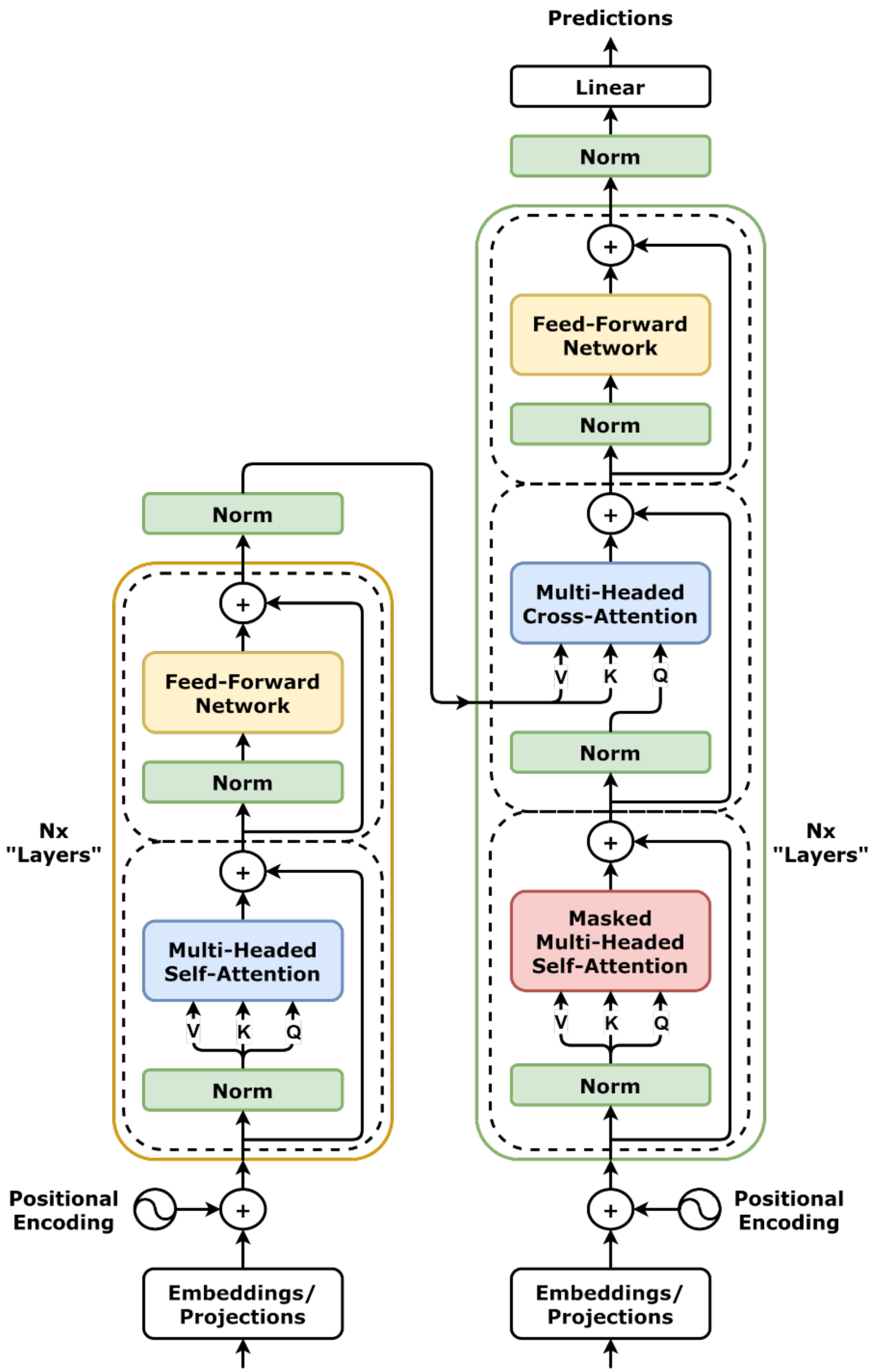

Session 18

Transformer

Reference

[1] Bishop, Christopher M., and Hugh Bishop. Deep learning: Foundations and concepts. Springer Nature, 2023.

[2] By dvgodoy - https://github.com/dvgodoy/dl-visuals/?tab=readme-ov-file

[3] https://arxiv.org/abs/1706.03762

https://physics.ipm.ac.ir/~ansarifard/

پژوهشگاه دانش‌های بنیادی (مرکز تحقیقات فیزیک نظری و ریاضیات)

IPM

INSTITUTE FOR RESEARCH IN FUNDAMENTAL SCIENCES

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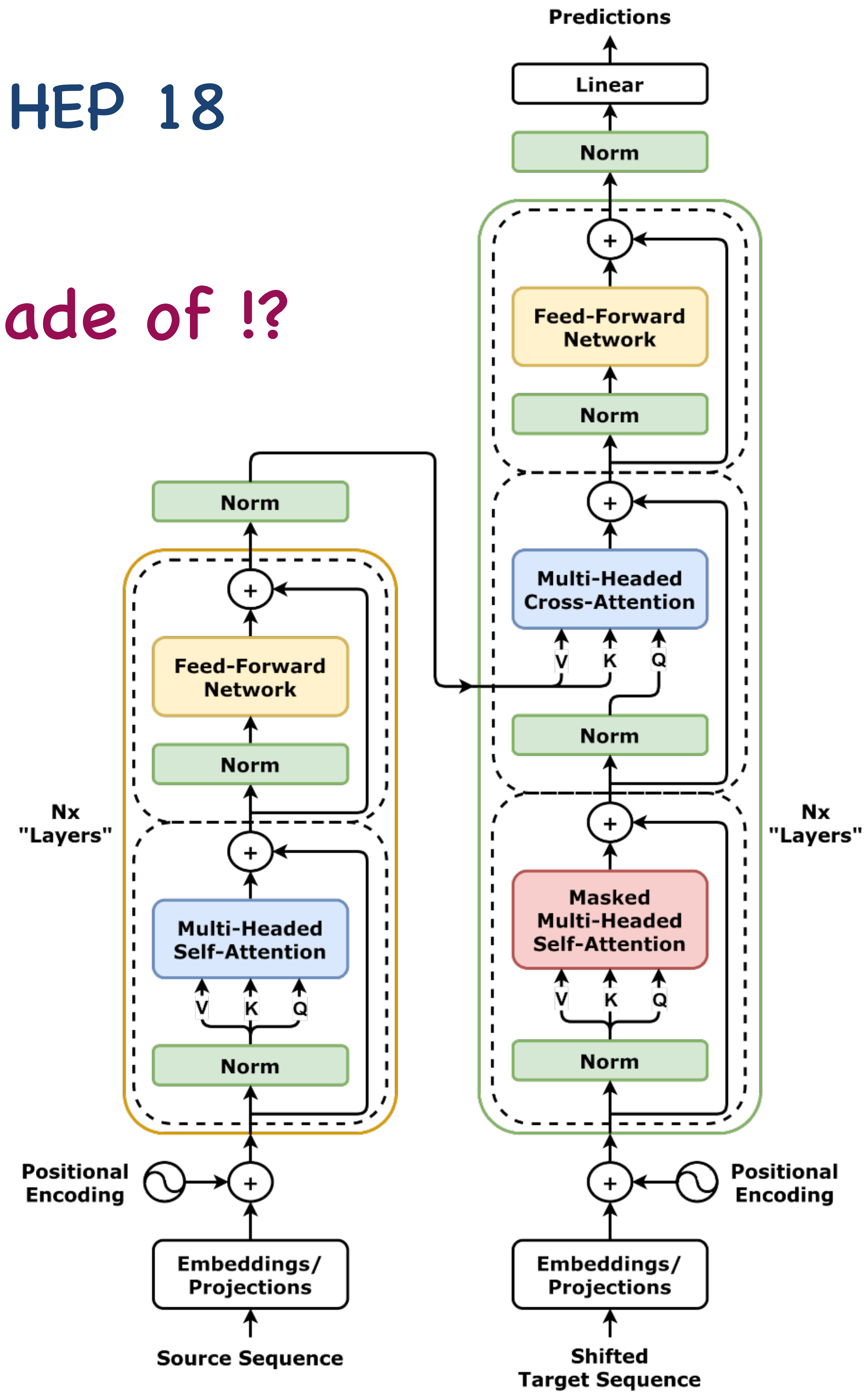
What Transformer is made of !?



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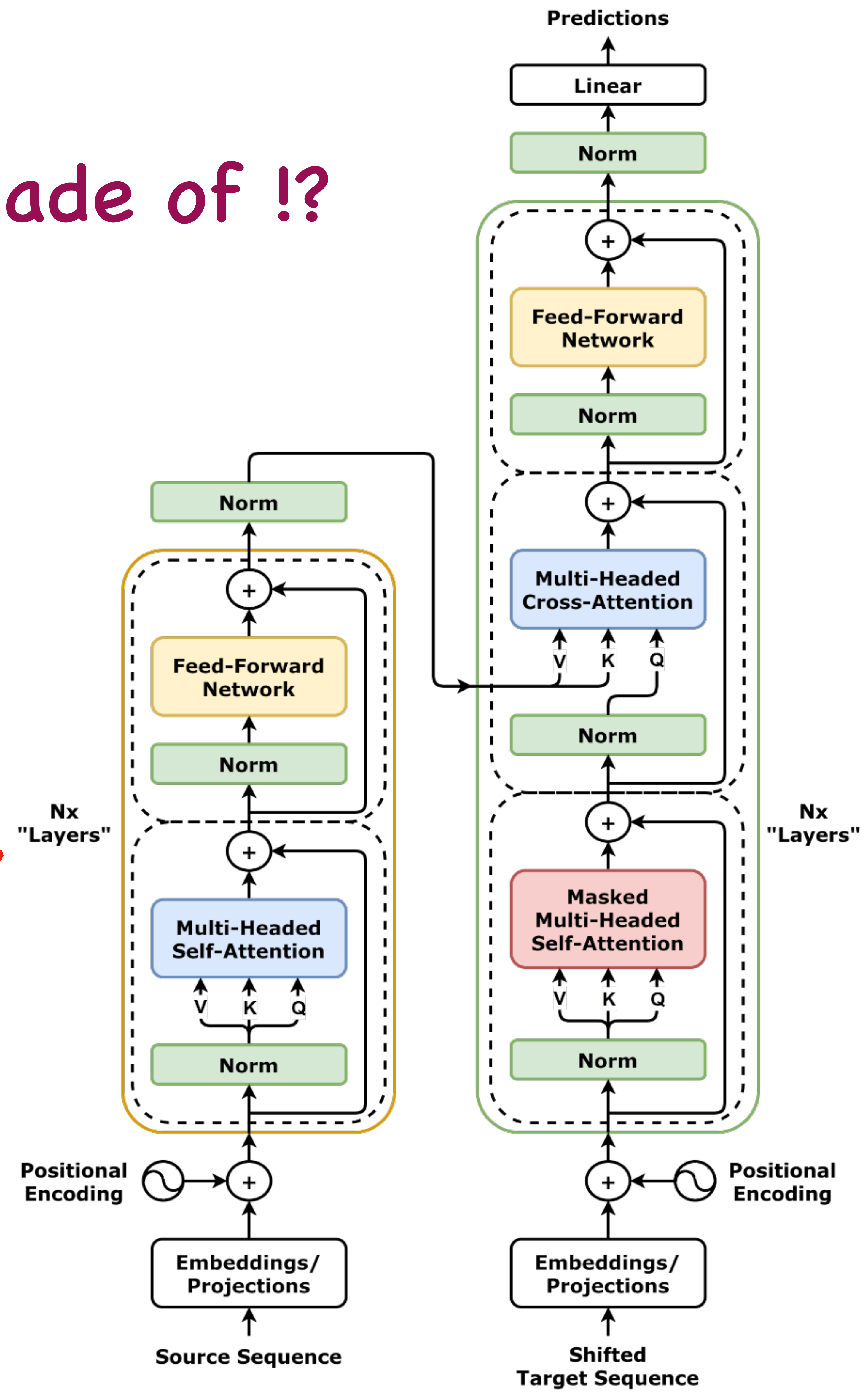
What Transformer is made of !?

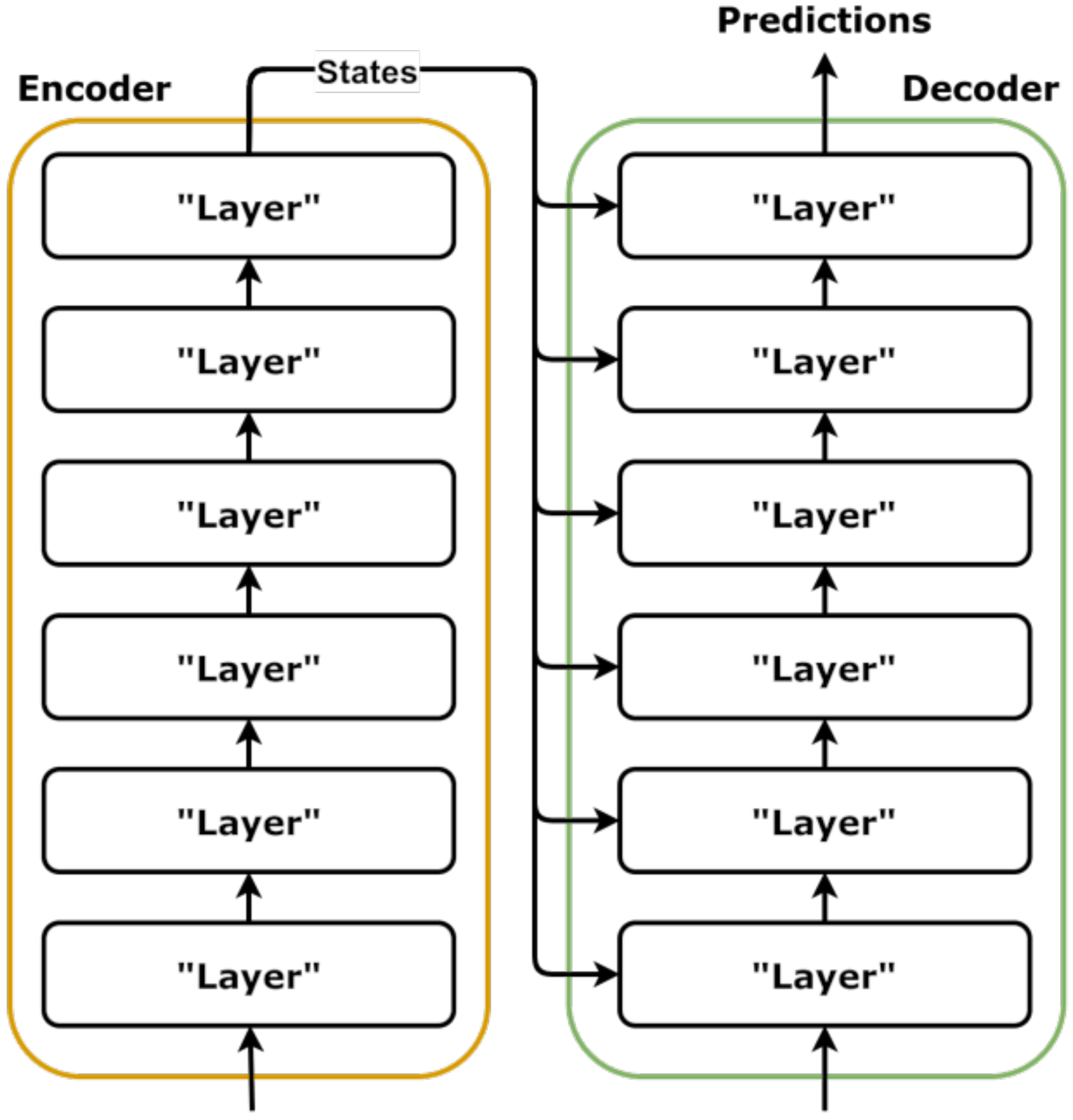
Stacked Layers

Early model ~ 6

Small models ~ 12

Large models ~ 100





(a) Stacked "Layers"

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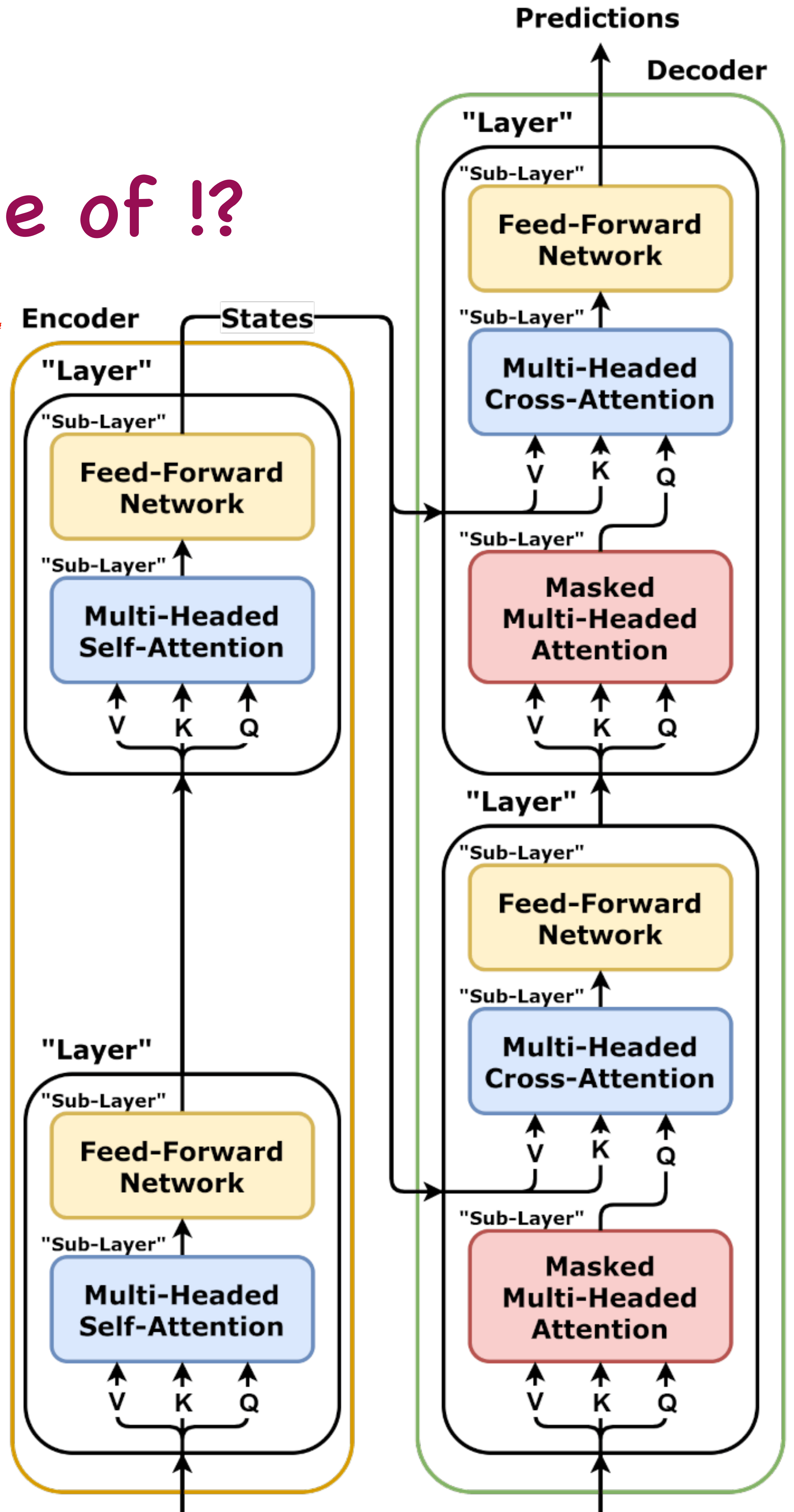
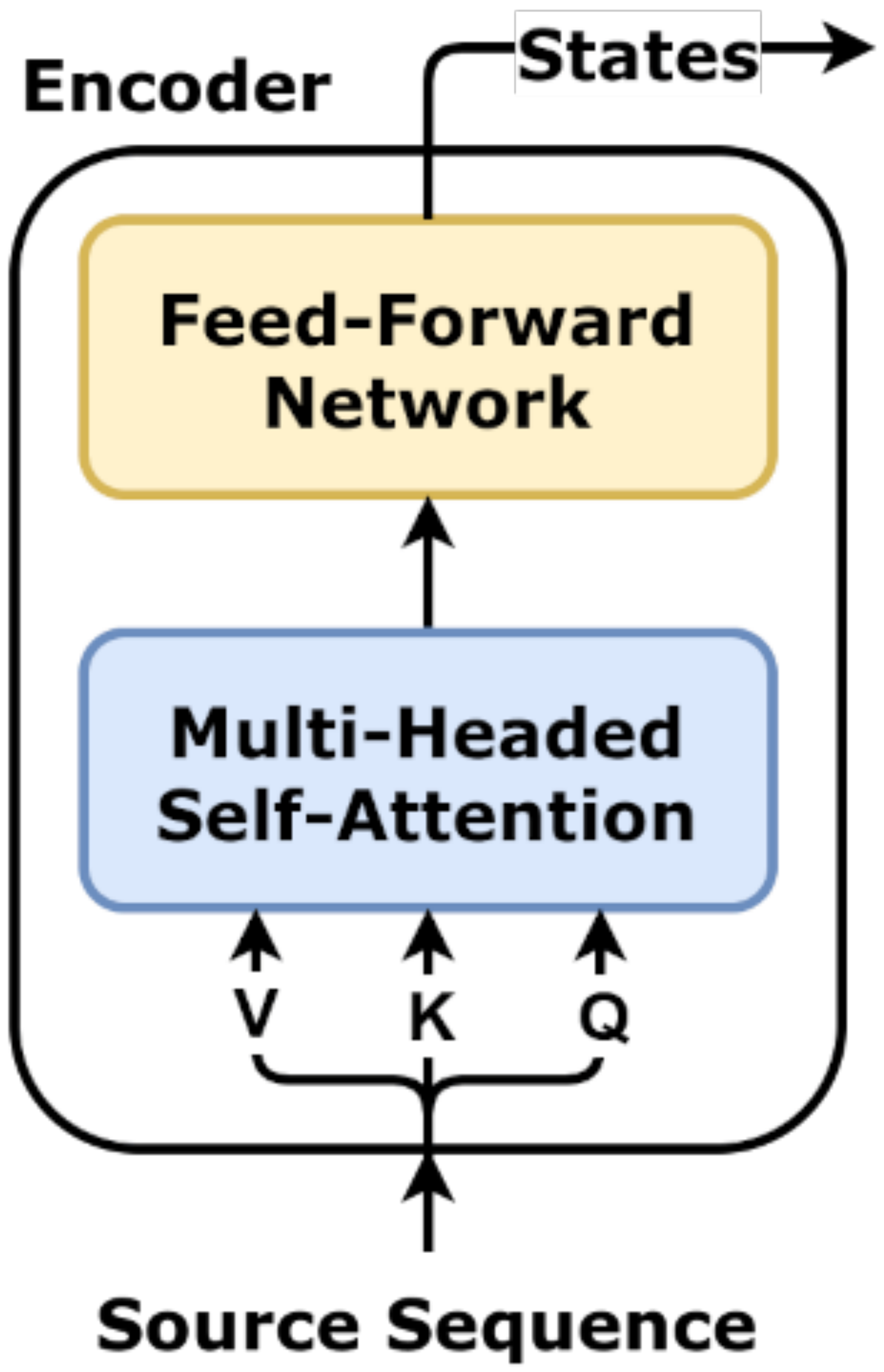
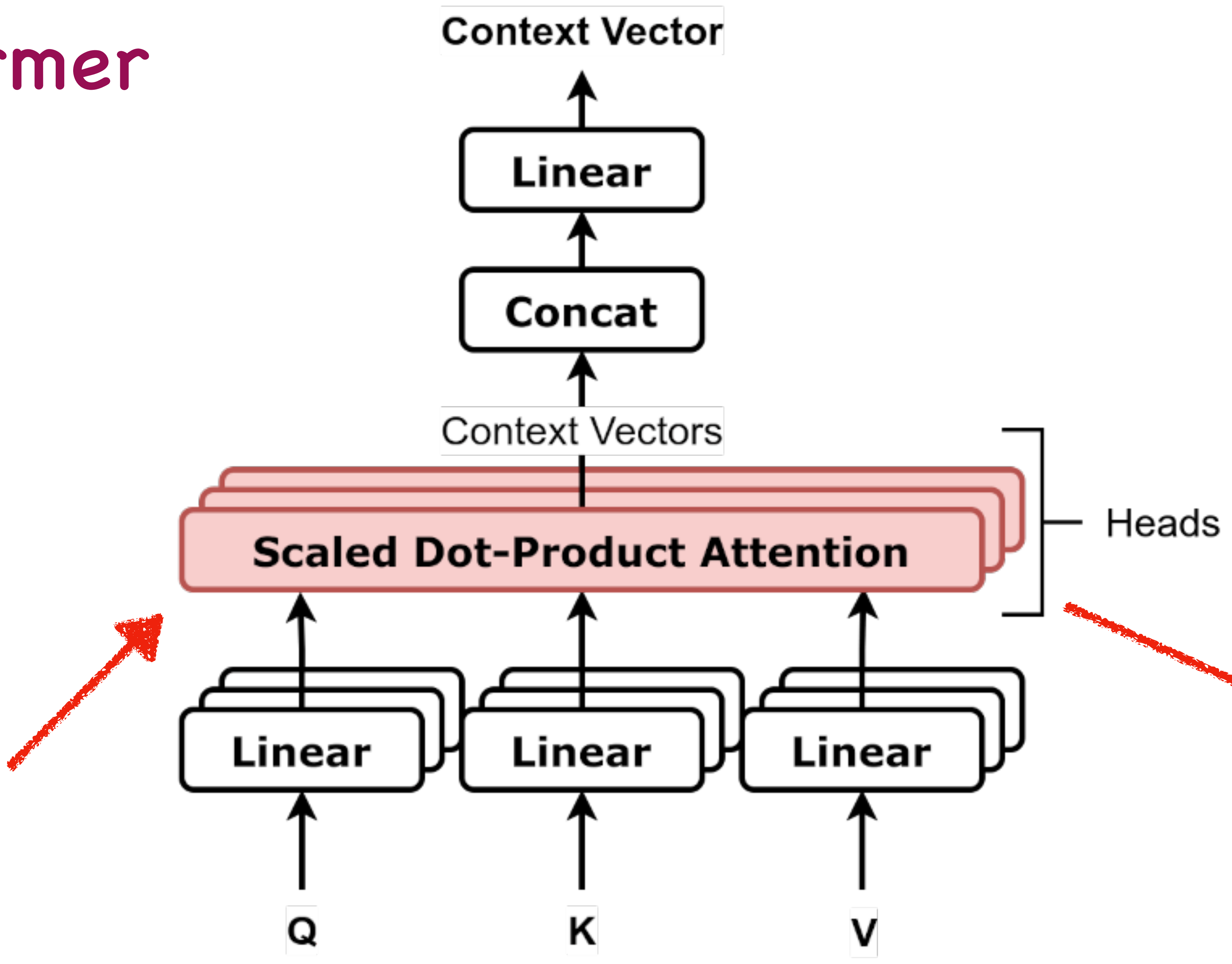
What Transformer is made of !?

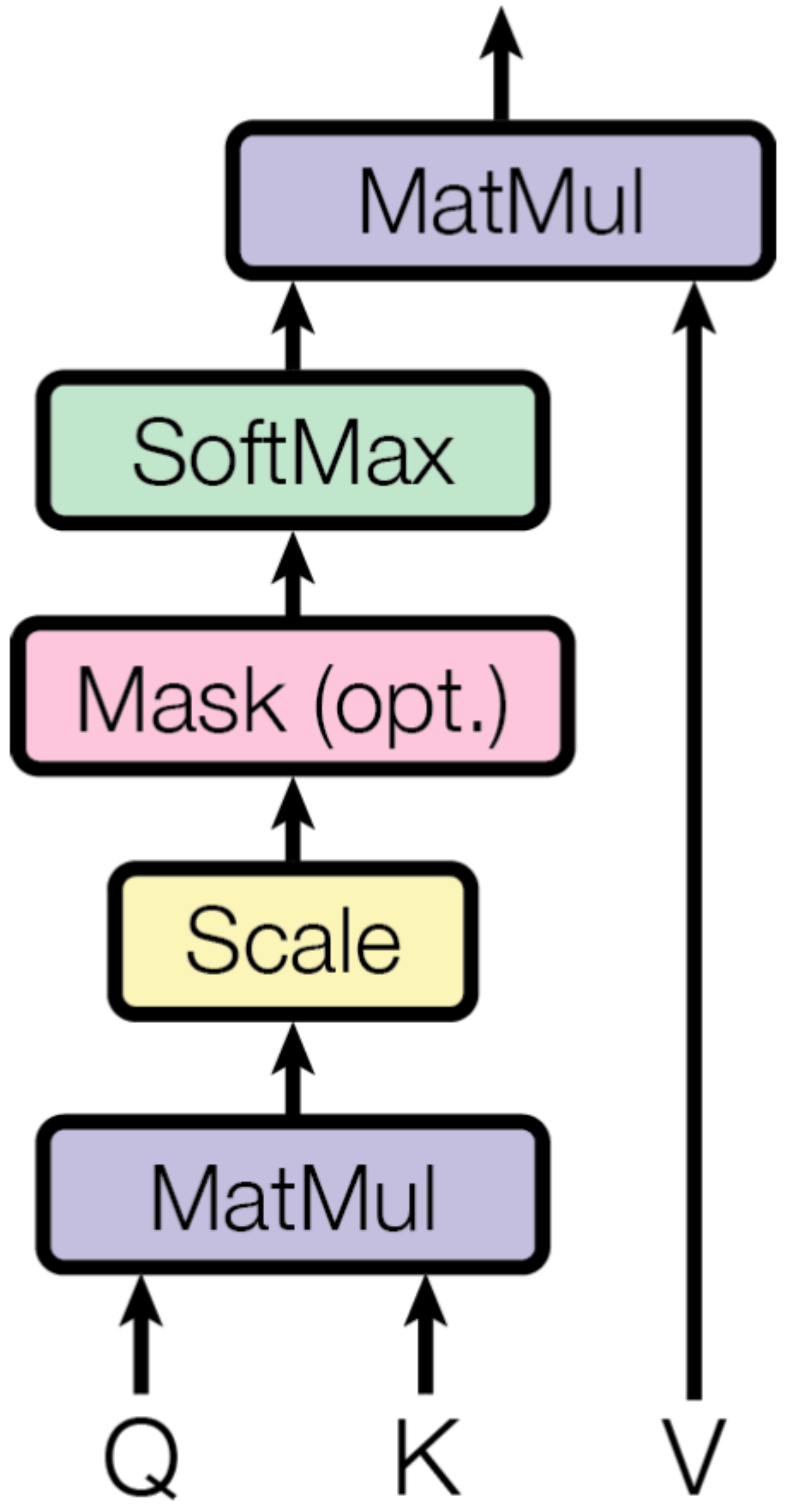
Encoder and Decoder blocks



Encoder Transformer

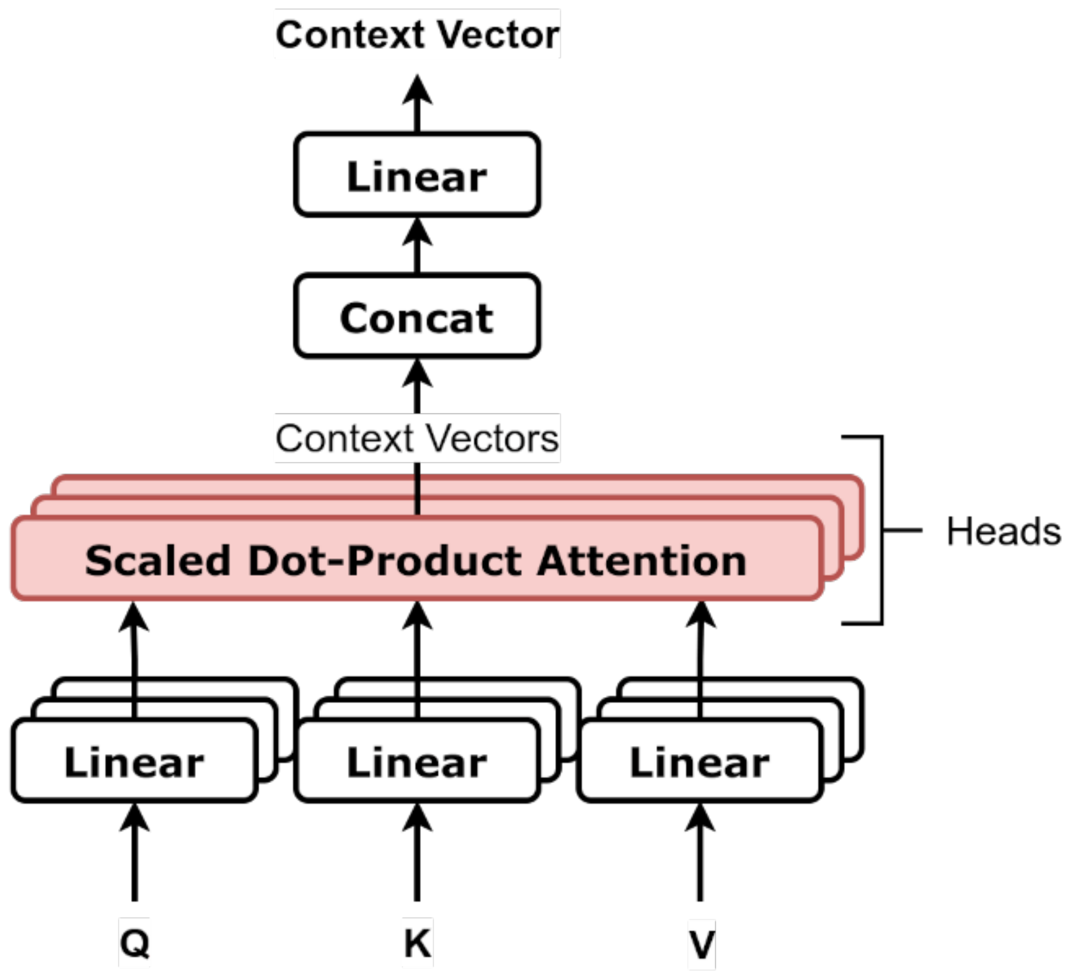






Attention Head

Dic = vocabulary size (Ex:: Dic = 10,000)



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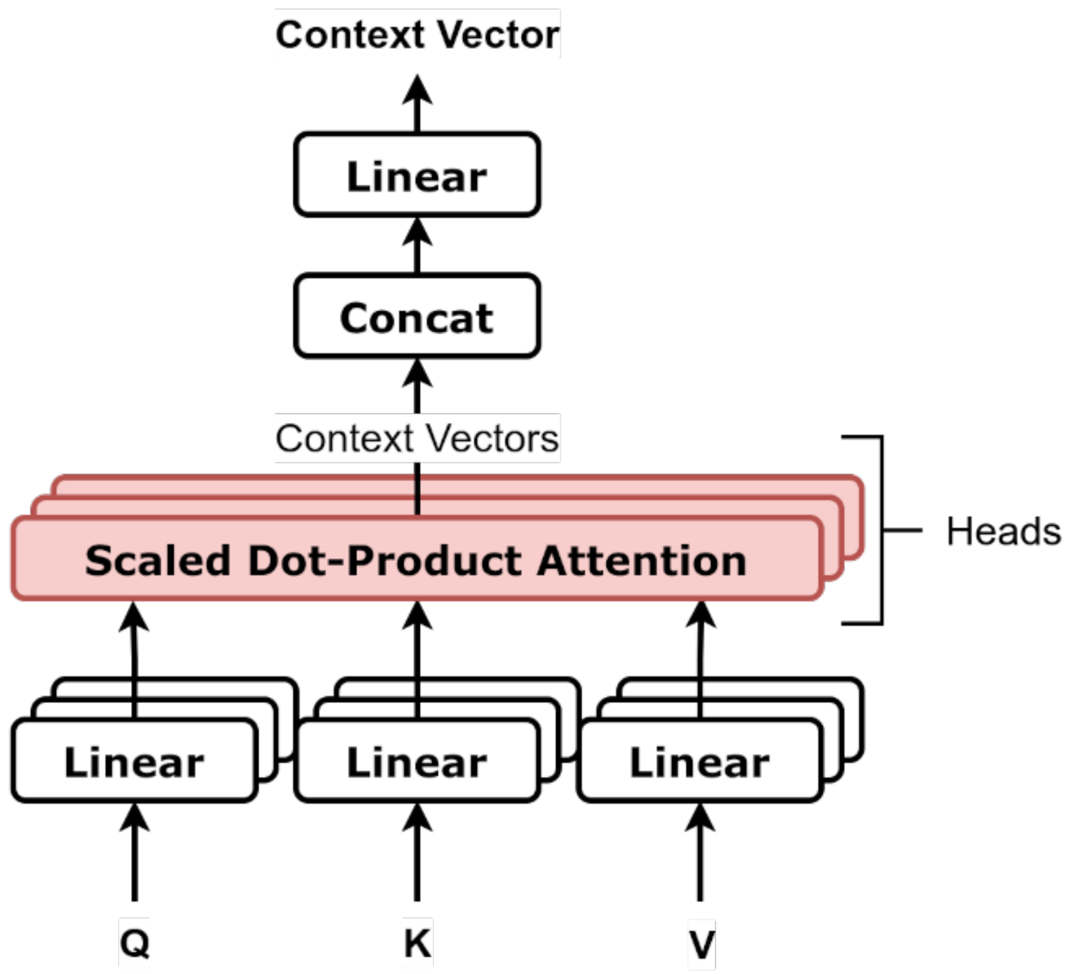

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N = number of input tokens (sequence length) (Ex:: N = 4)

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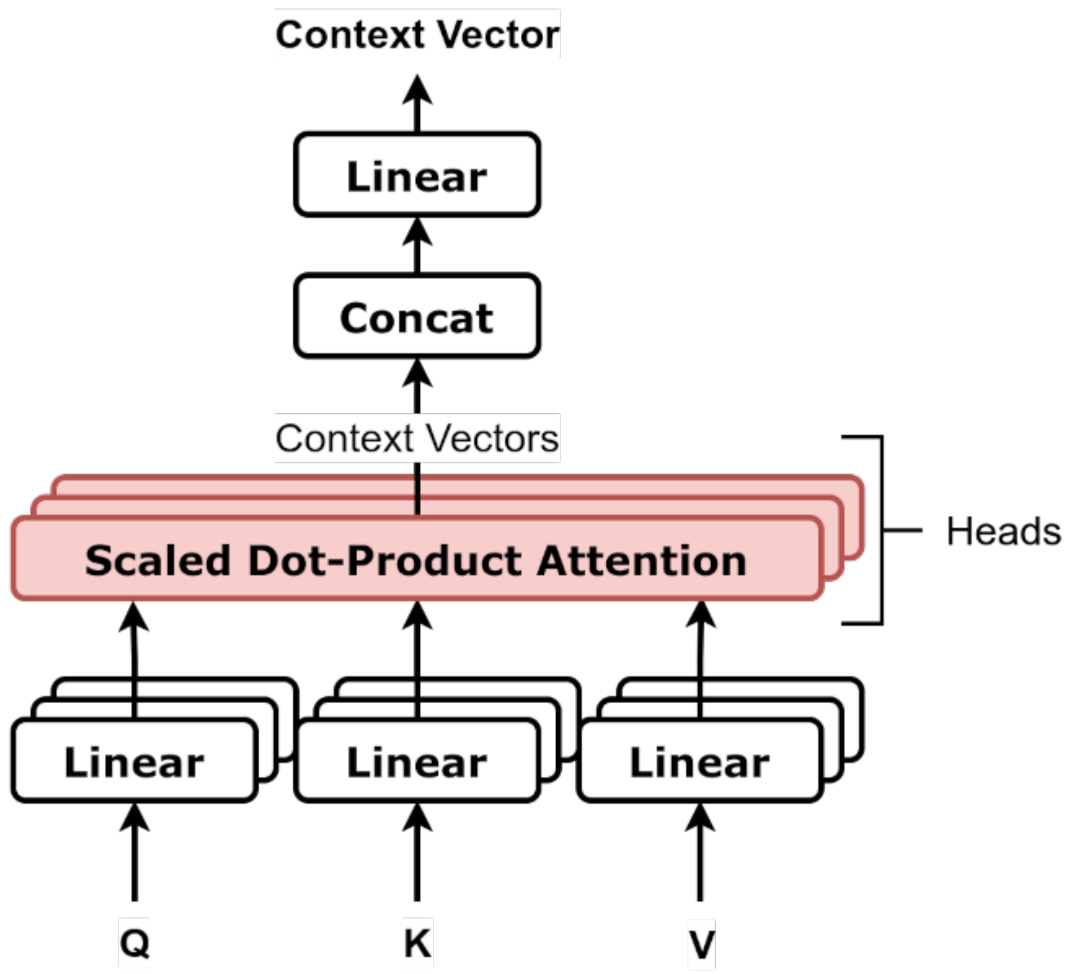

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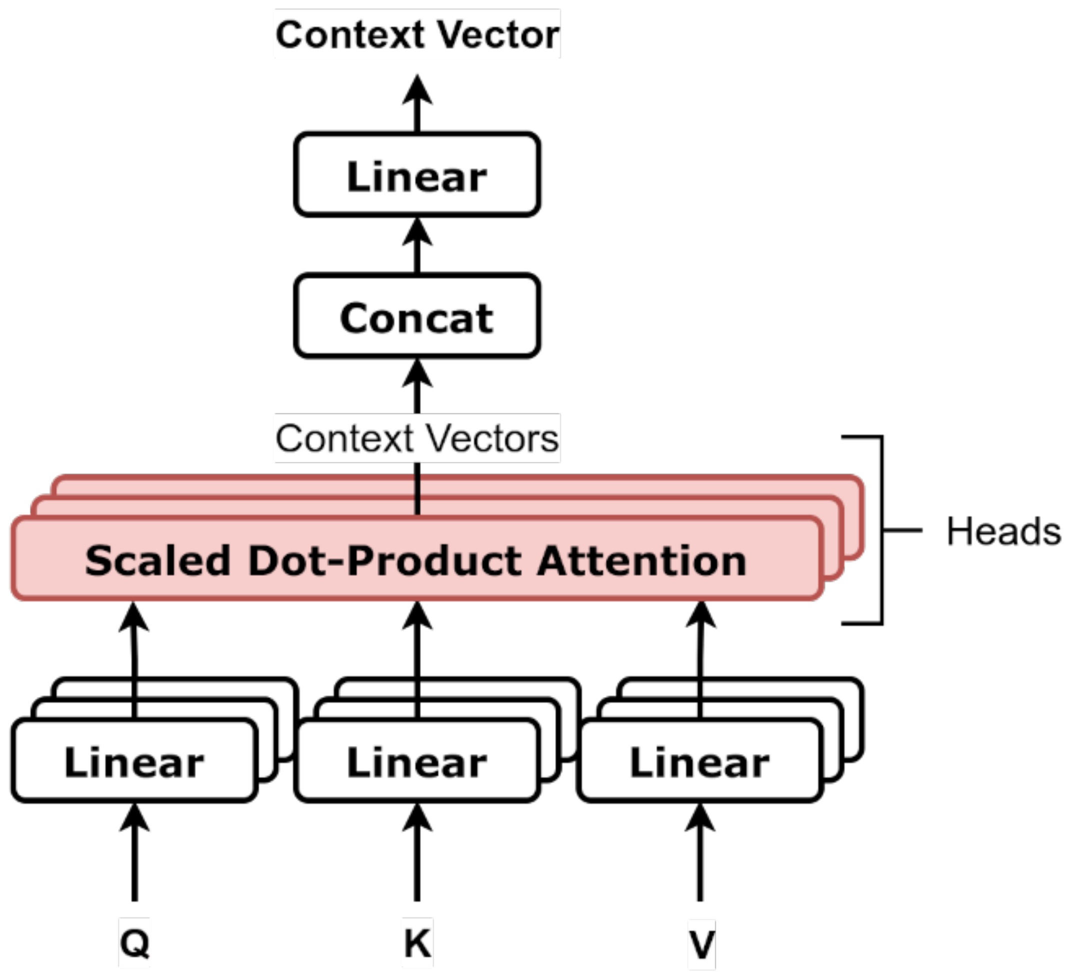

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Assign number to building block of the semantic word

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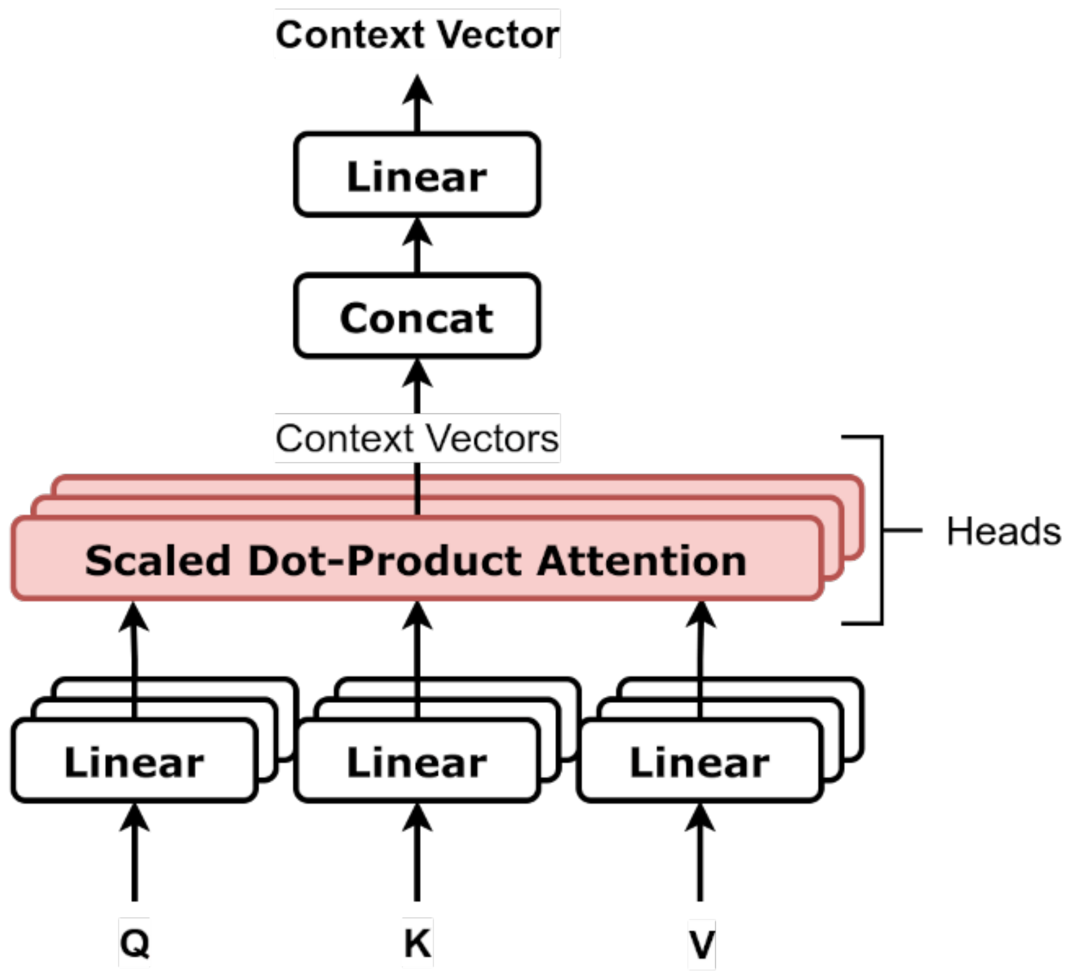

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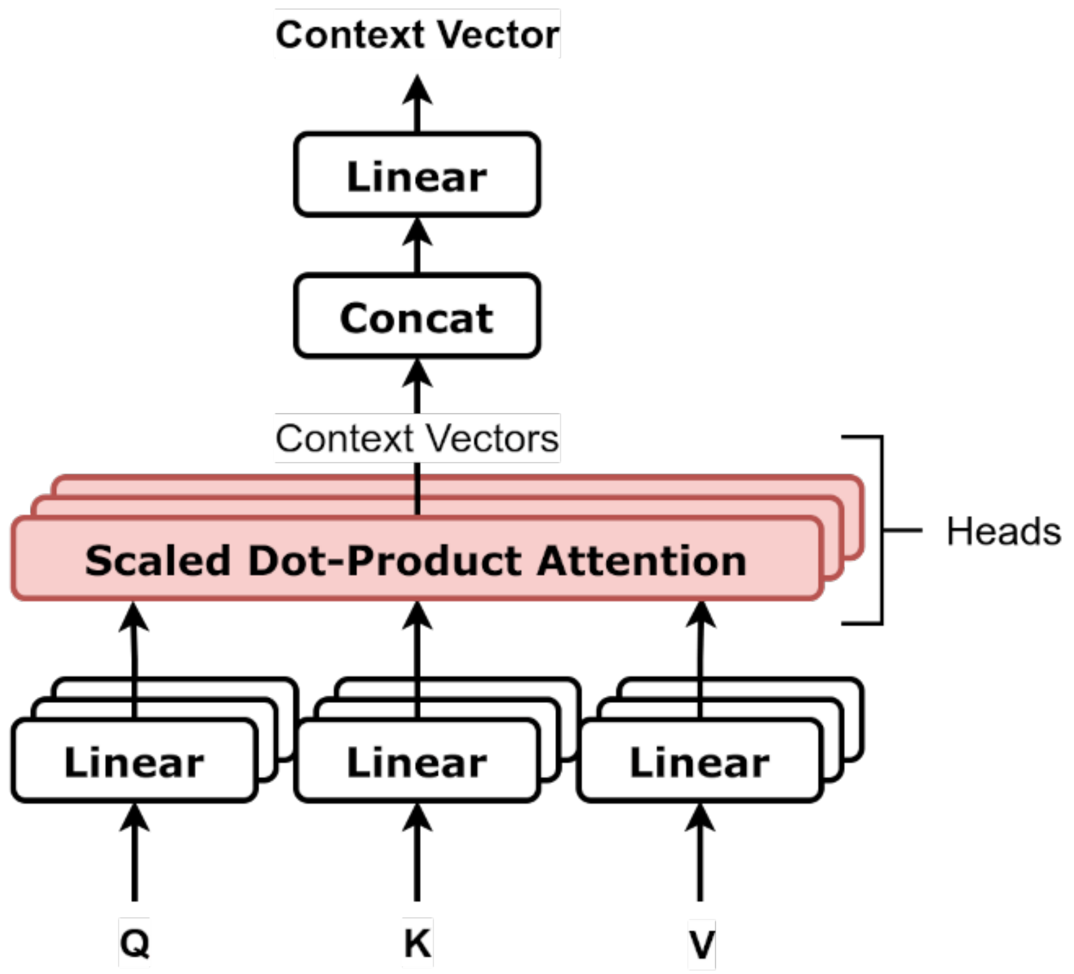
In natural language representation it reduce to choose between character level or word level

Attention Head



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D_model = model hidden dimension (embedding dimension) (Ex:: D_model = 8)

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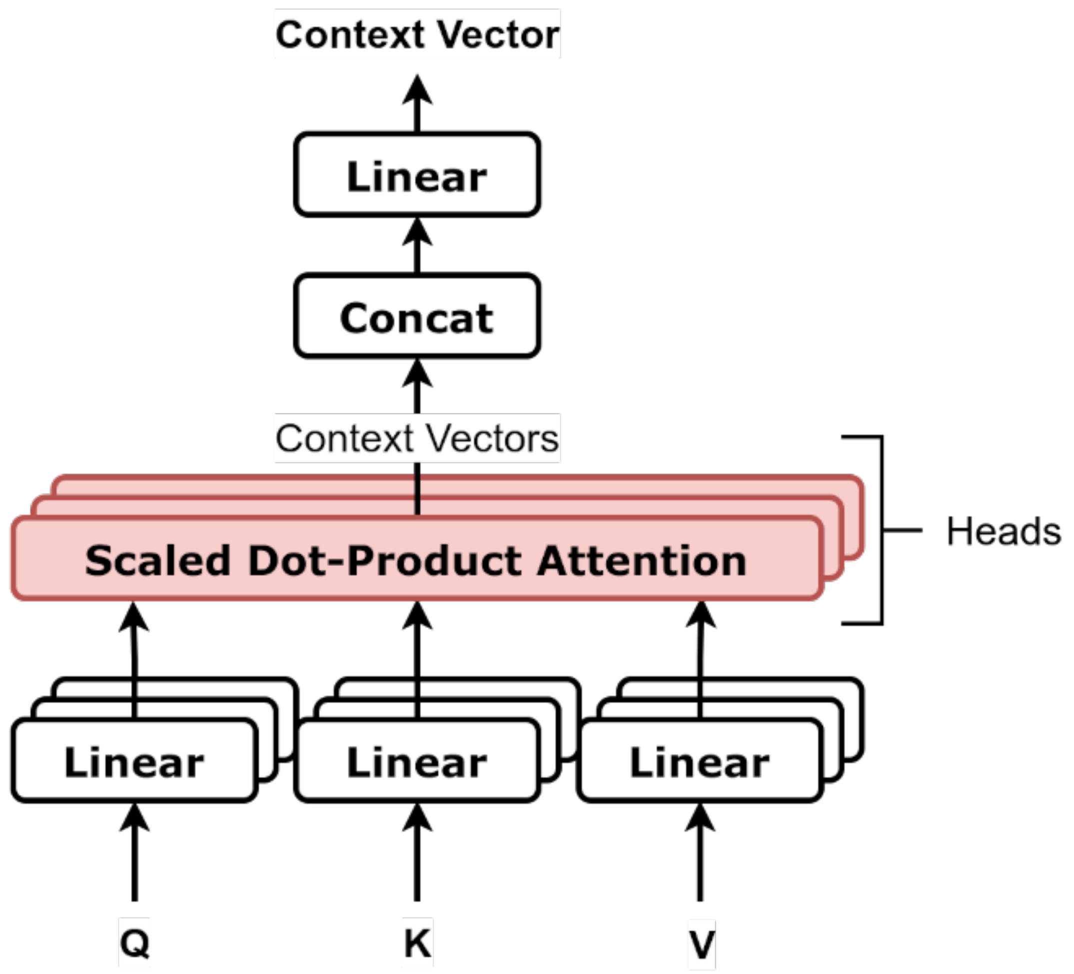

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One-hot vectors : x1= [0,...,0,1,0,...,0]

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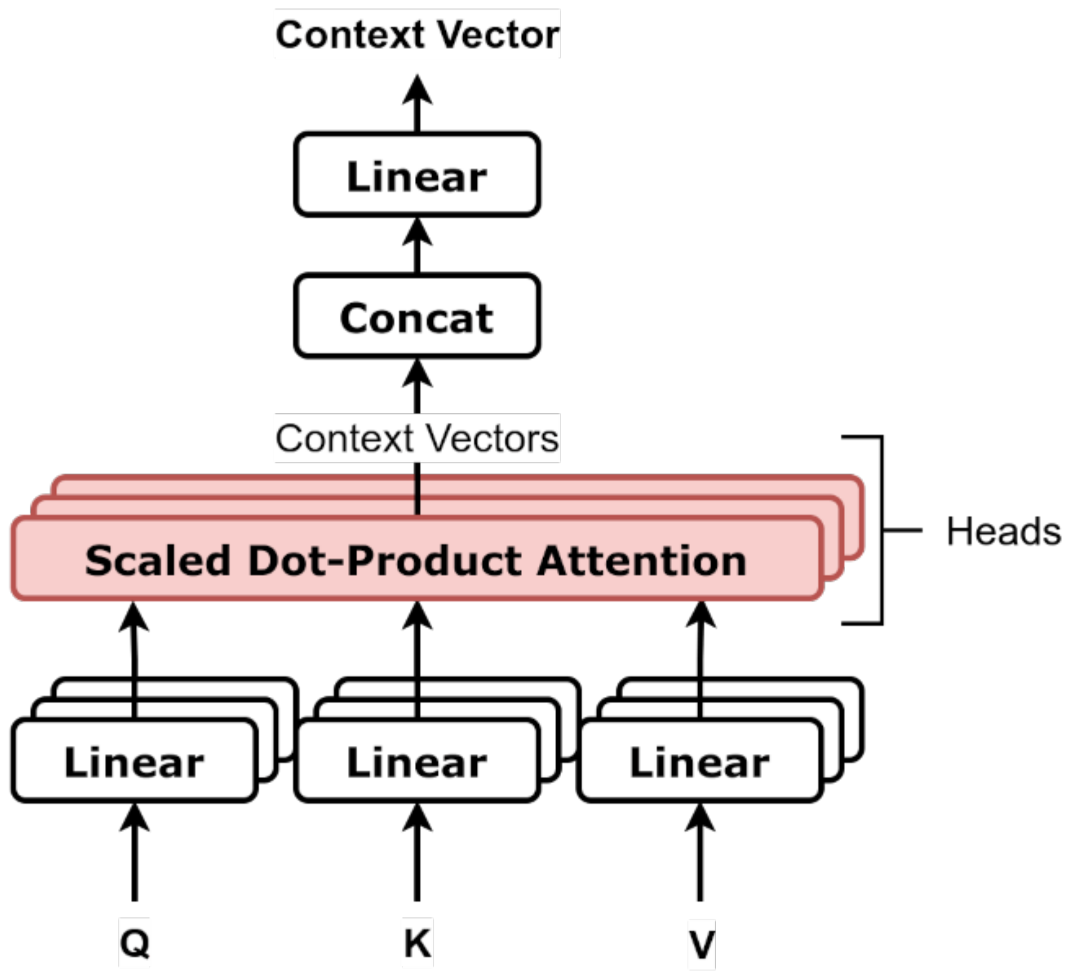

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Embedding with E which is a matrix of shape (Dic, D_model), Example : $x1 \rightarrow E^{\top}x1$

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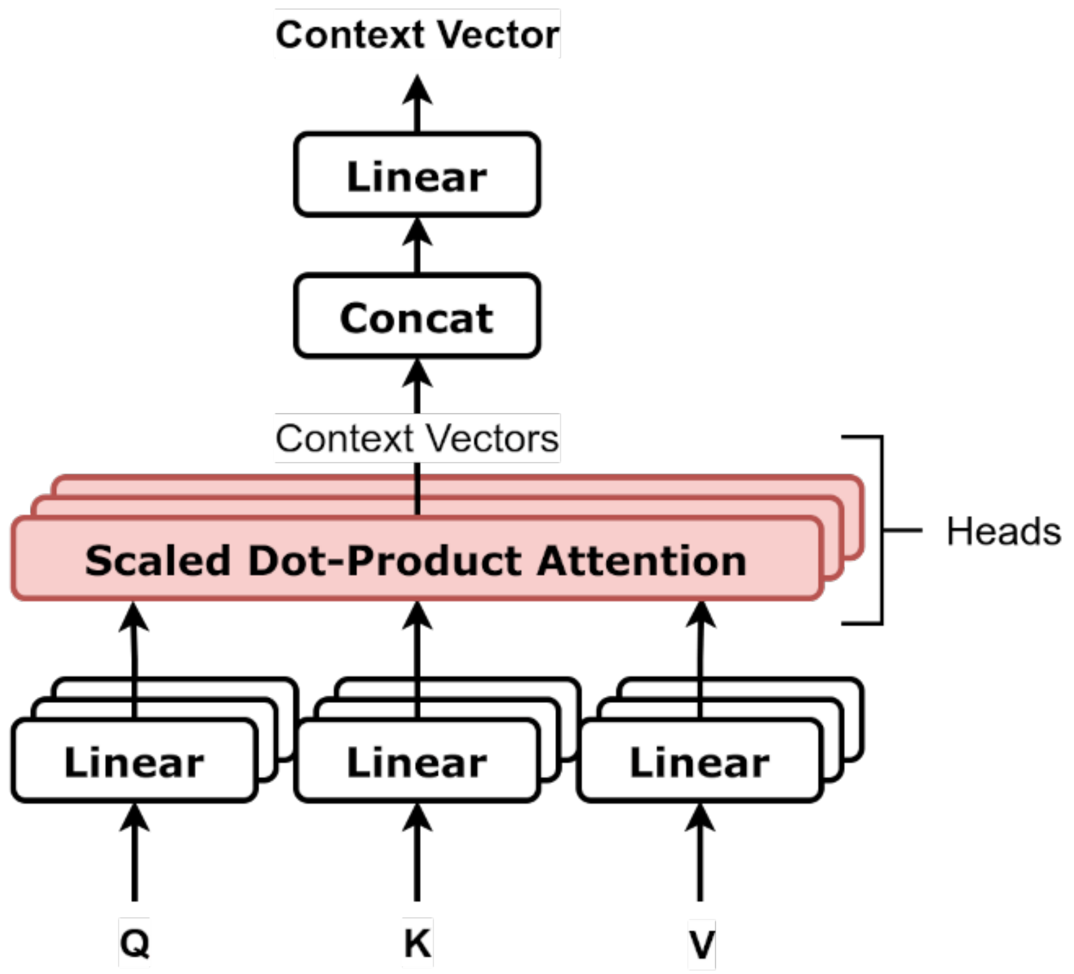

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starts with random values

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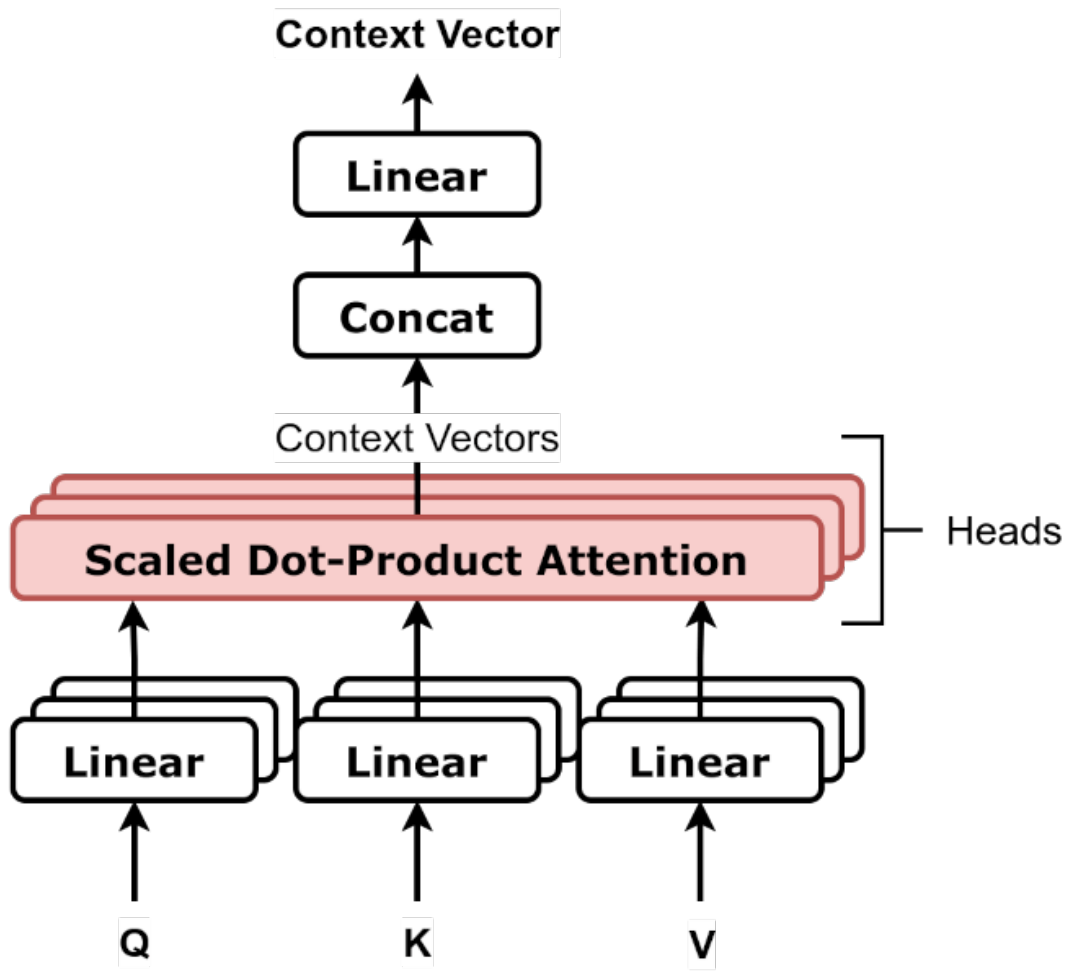

X : (N, D_model)

Dic : total dictionary

$$Q = XW^q \quad K = XW^k \quad V = XW^v$$

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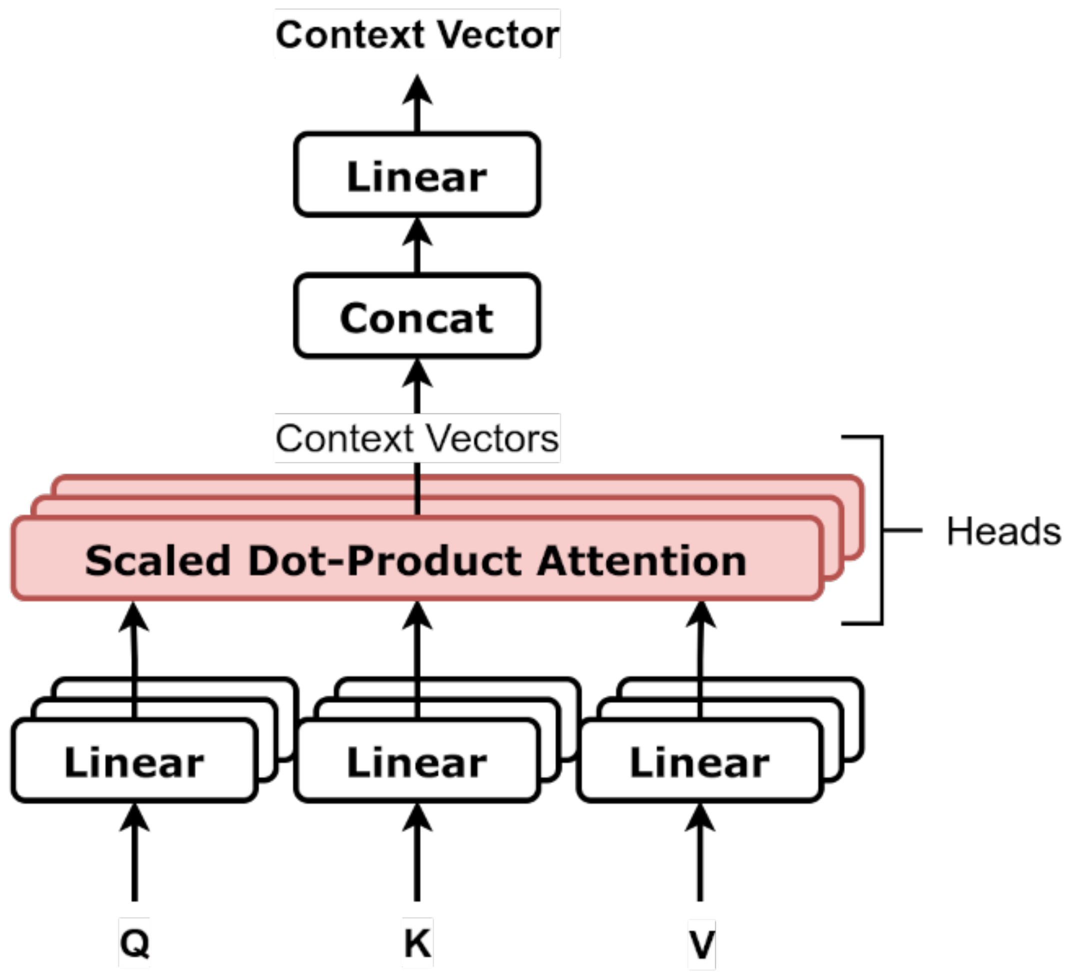

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$$\begin{bmatrix} x_{11} & x_{12} & x_{13} & x_{14} & x_{15} & x_{16} & x_{17} & x_{18} \\ x_{21} & x_{22} & x_{23} & x_{24} & x_{25} & x_{26} & x_{27} & x_{28} \\ x_{31} & x_{32} & x_{33} & x_{34} & x_{35} & x_{36} & x_{37} & x_{38} \\ x_{41} & x_{42} & x_{43} & x_{44} & x_{45} & x_{46} & x_{47} & x_{48} \end{bmatrix}$$

AI-Driven HEP 18

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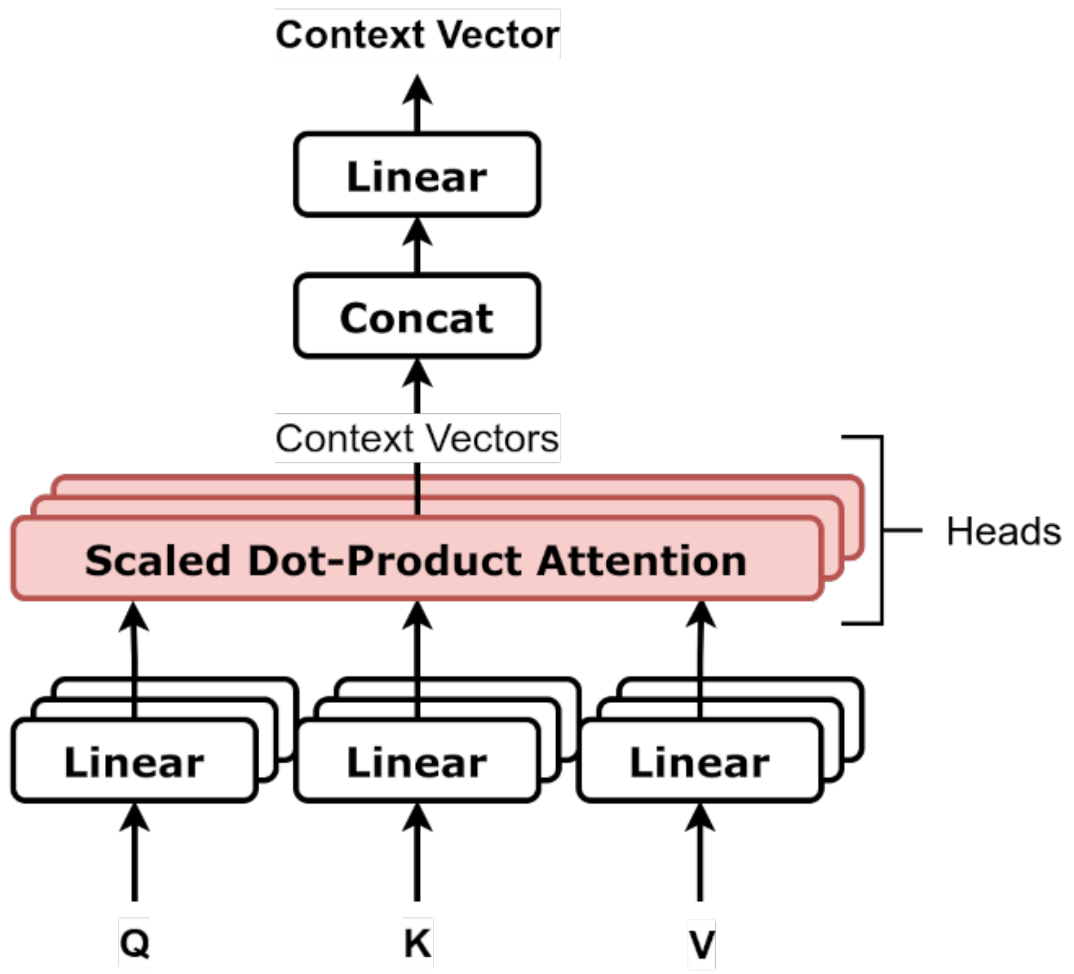

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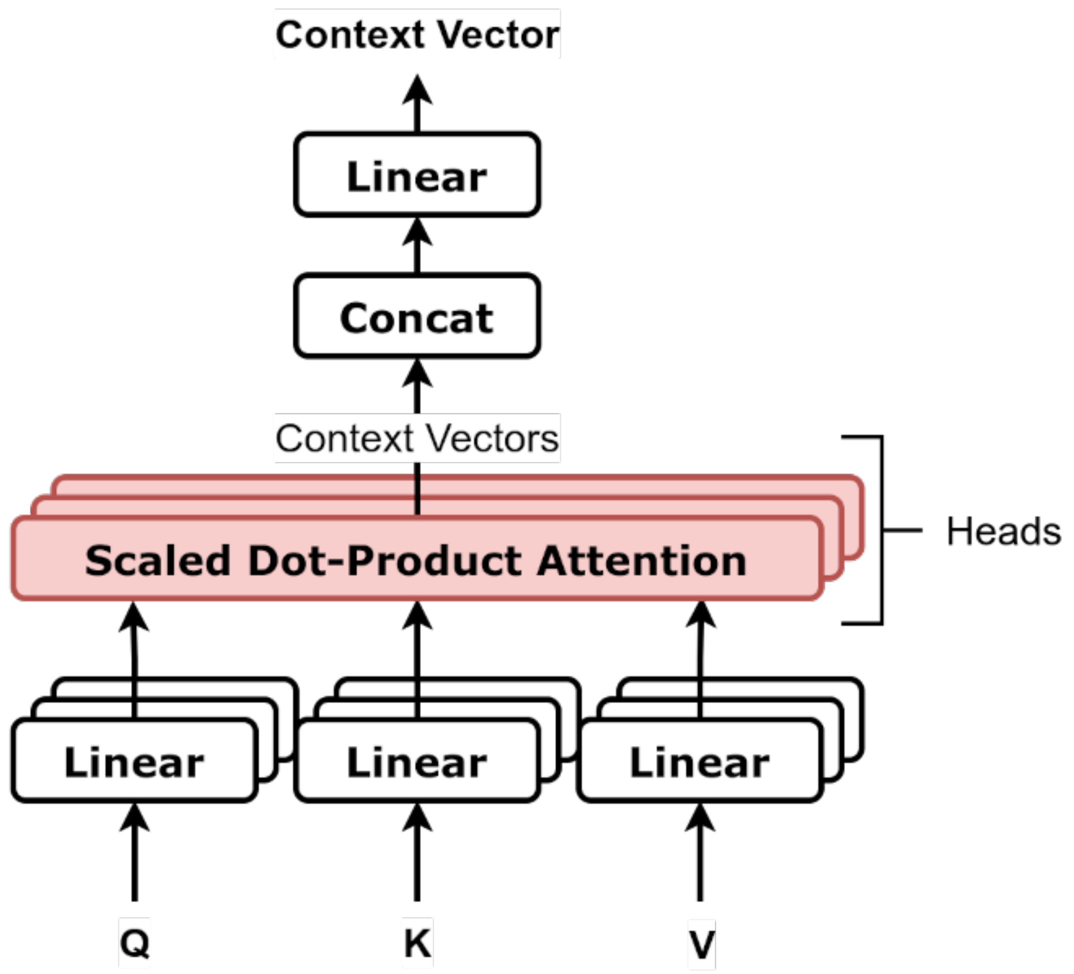

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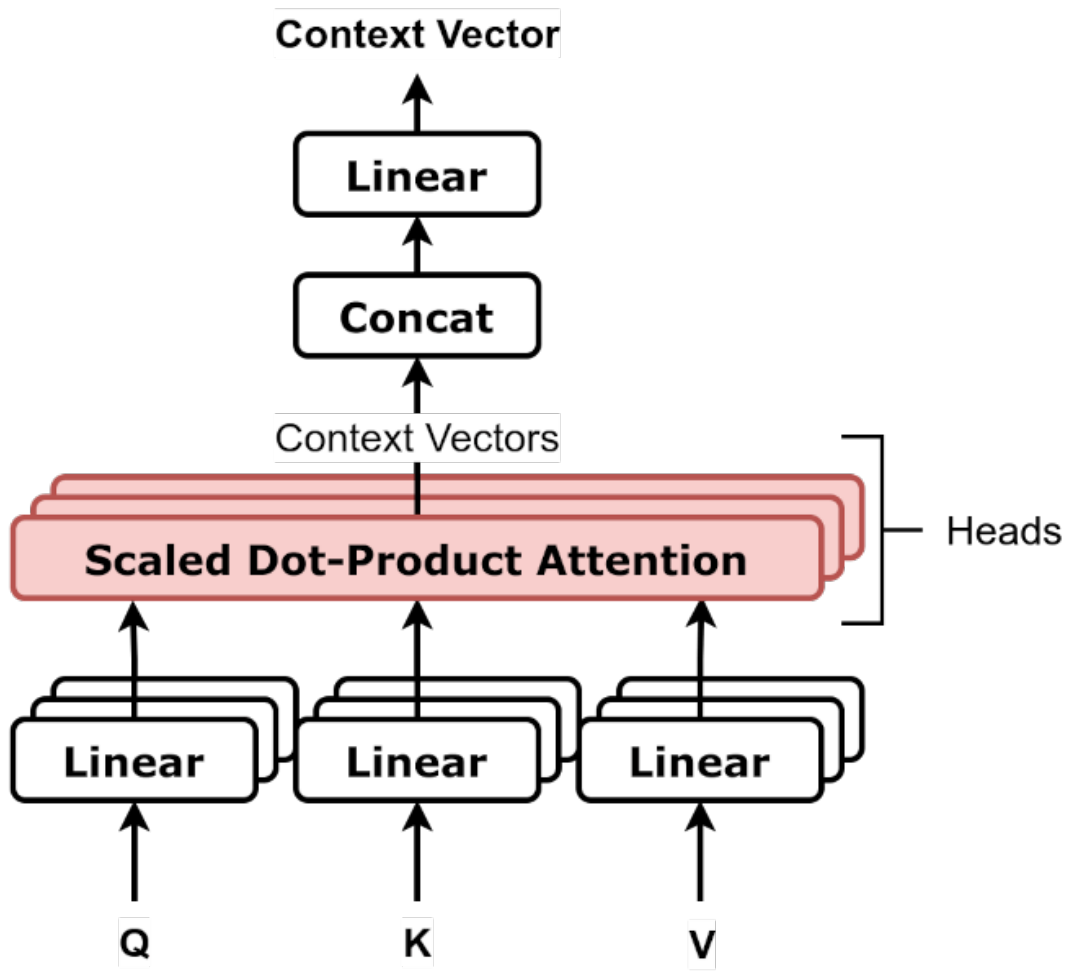
Similar to a single linear node in FCNN

Attention Head



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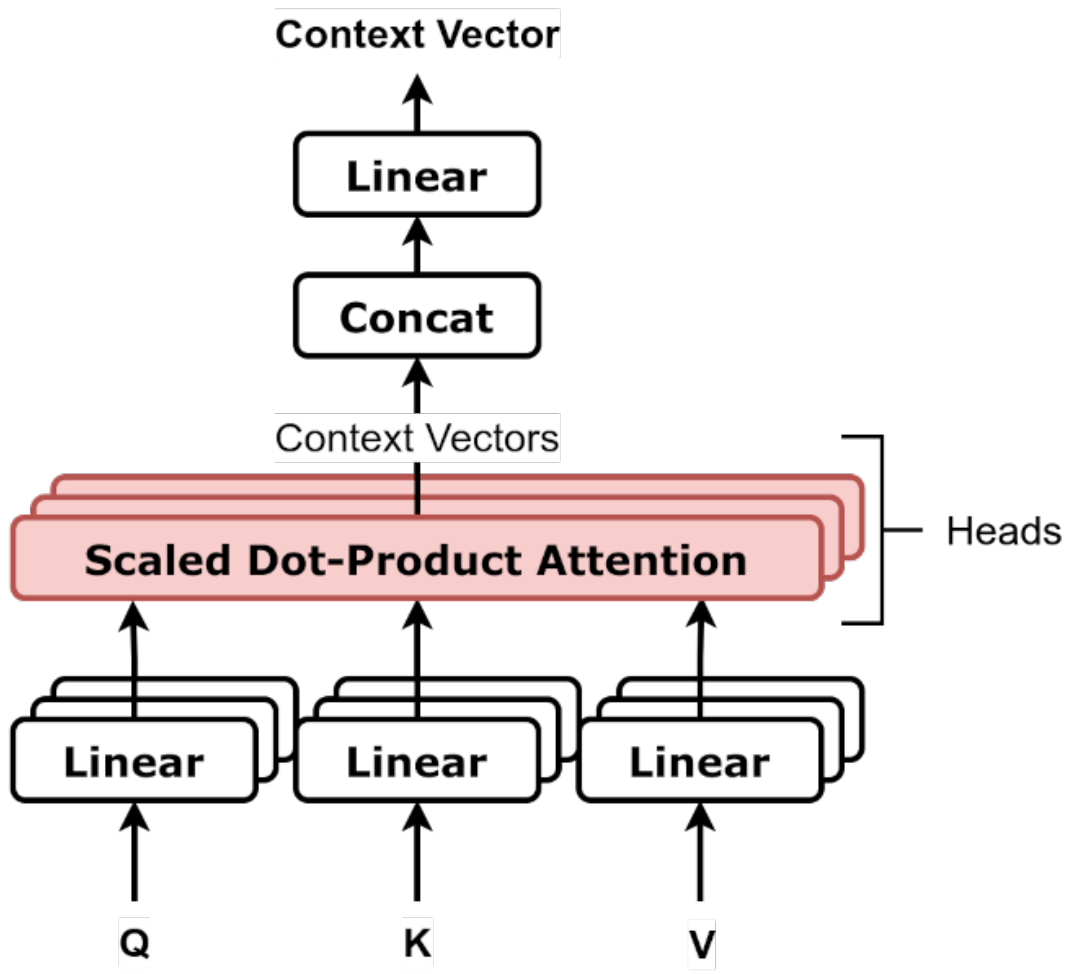
h = number of attention heads (Ex:: h = 4)

D_head = D_model / h = head dimension.

With our numbers: D_head = 8/4 = 2

Attention Head



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$$Q_h = QW_h^q$$

$$\begin{bmatrix} q_{11} & q_{12} & q_{13} & q_{14} & q_{15} & q_{16} & q_{17} & q_{18} \\ q_{21} & q_{22} & q_{23} & q_{24} & q_{25} & q_{26} & q_{27} & q_{28} \\ q_{31} & q_{32} & q_{33} & q_{34} & q_{35} & q_{36} & q_{37} & q_{38} \\ q_{41} & q_{42} & q_{43} & q_{44} & q_{45} & q_{46} & q_{47} & q_{48} \end{bmatrix}$$

This projection matrix is not learnable

$$\begin{bmatrix} w_{11}^q & w_{12}^q \\ w_{21}^q & w_{22}^q \\ w_{31}^q & w_{32}^q \\ w_{41}^q & w_{42}^q \\ w_{51}^q & w_{52}^q \\ w_{61}^q & w_{62}^q \\ w_{71}^q & w_{72}^q \\ w_{81}^q & w_{82}^q \end{bmatrix}$$

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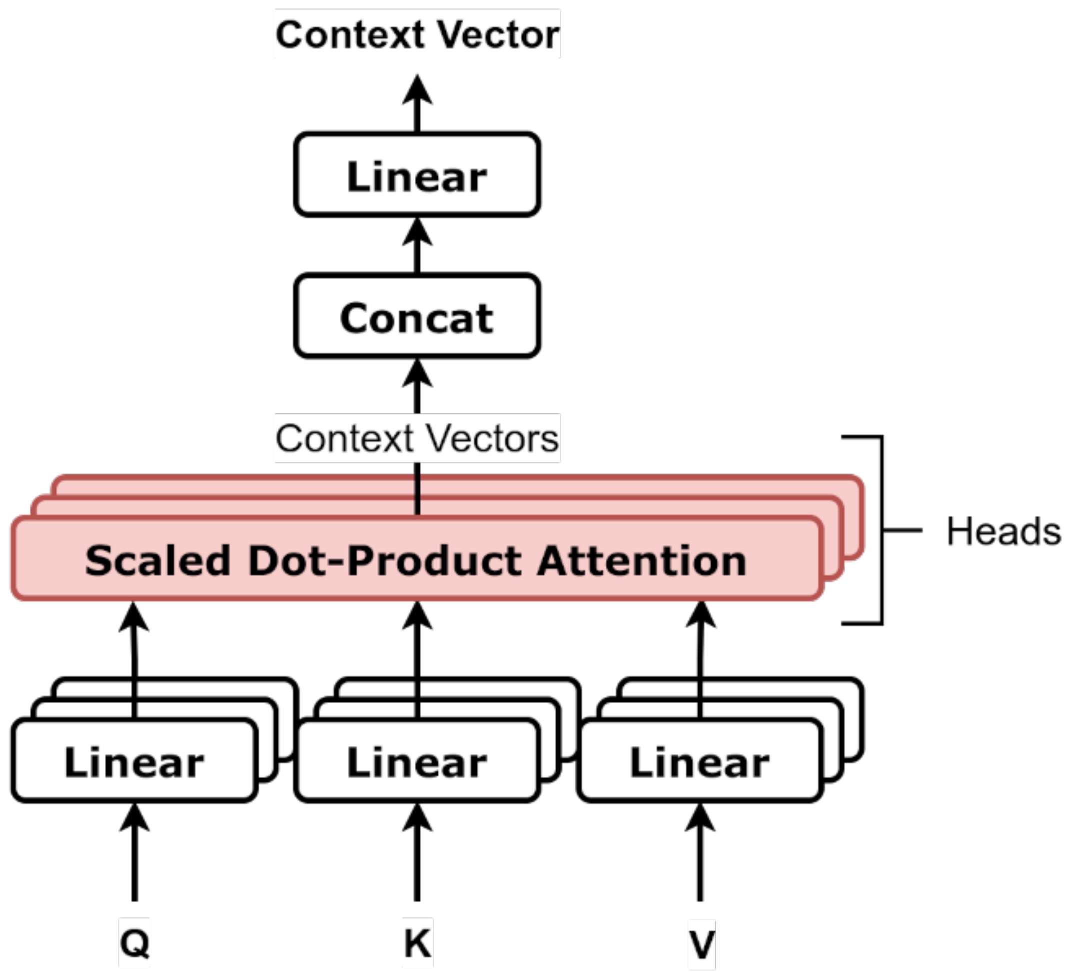
Attention Head



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$$Q = XW^q \qquad K = XW^k \qquad V = XW^v$$

$$X = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{bmatrix} \in \mathbb{R}^{2\times 4}$$

$$W_1^Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad W_2^Q = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$Q_2 = XW_2^Q = \begin{bmatrix} 3 & 4 \\ 7 & 8 \end{bmatrix}$$

$$W_Q = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \in \mathbb{R}^{4\times 4}$$

$$Q = XW_Q = X = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{bmatrix}$$

$$Q_1 = XW_1^Q = \begin{bmatrix} 1 & 2 \\ 5 & 6 \end{bmatrix}$$

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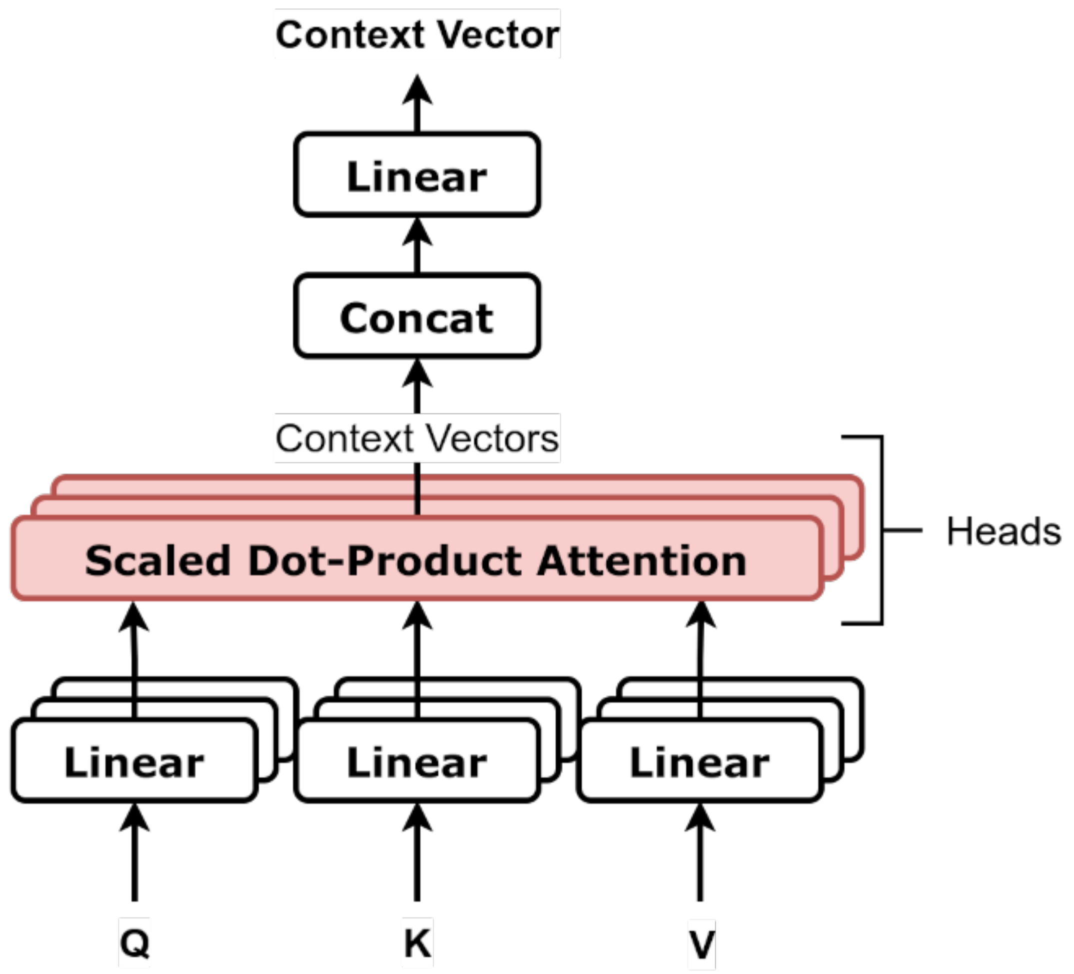

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Number of head	Number of D_model
Small ~ 8	Small ~ 512
Medium ~ 16	Medium ~ 1024
Large ~ 100	Large ~ 10000

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Attention Head

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$$Q_h = XW_h^q$$

$$K_h = XW_h^k$$

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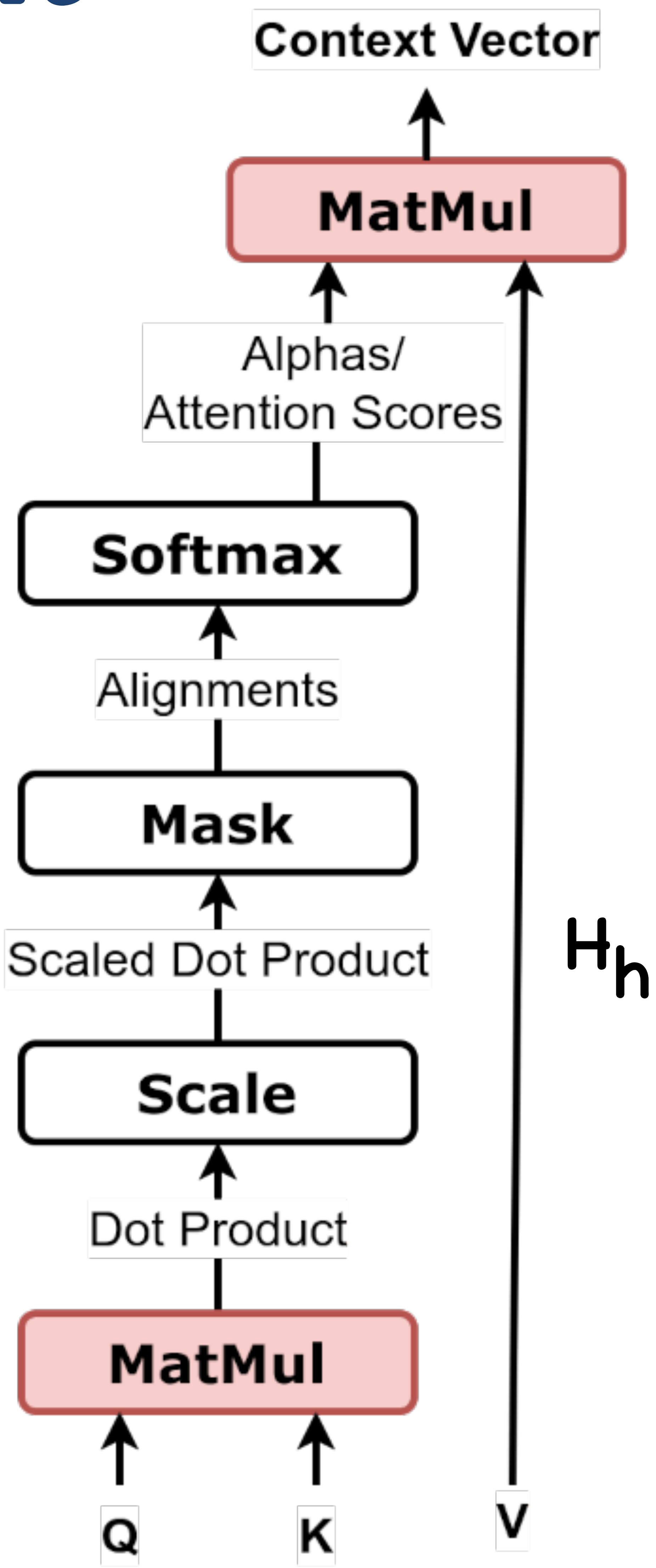

Scaled Dot-Product Attention

$$H_h = \text{Attention}(Q,K,V) = \text{SoftMax}\left(\frac{Q_h K_h^T}{\sqrt{D_{head}}}\right) V_h$$

variance of the dot product

Attention Head

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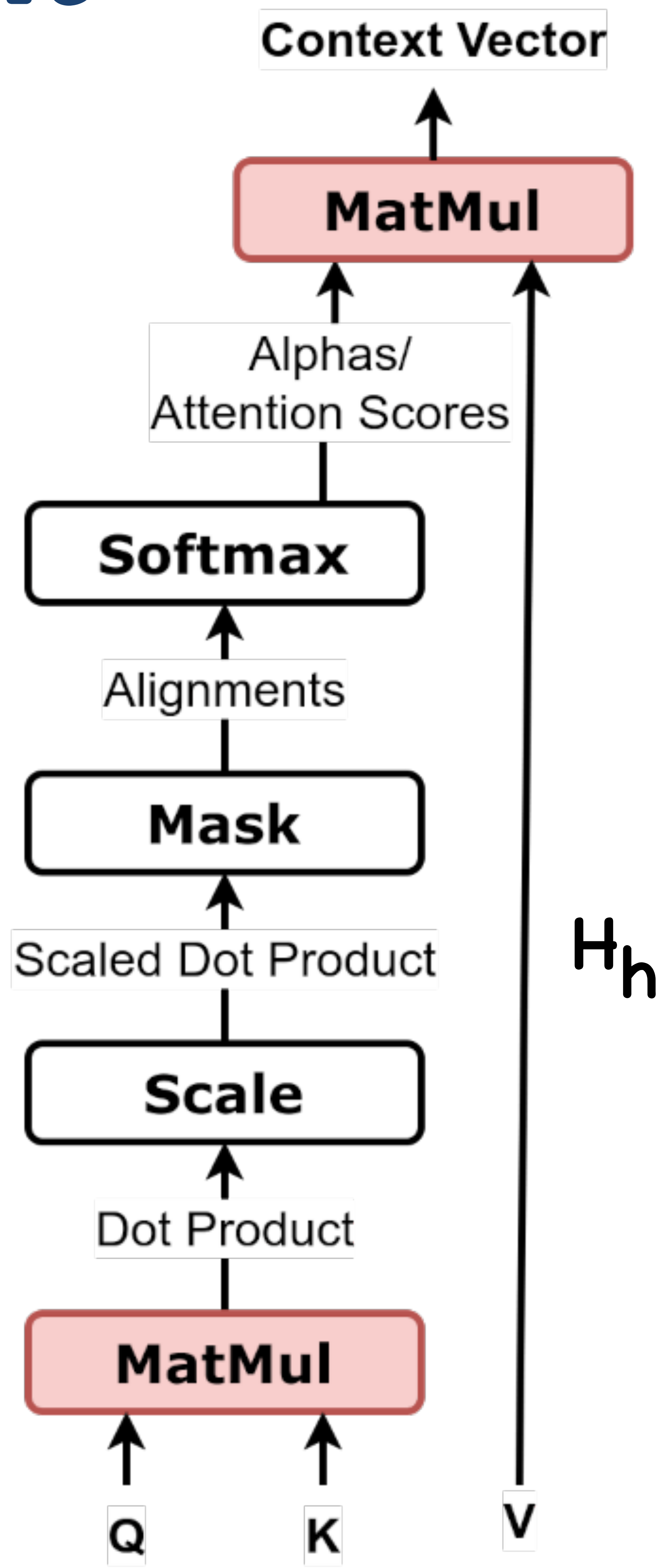

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variance of the dot product

$$Y = \text{Concat}(H_1,\ldots,H_h)W^O$$

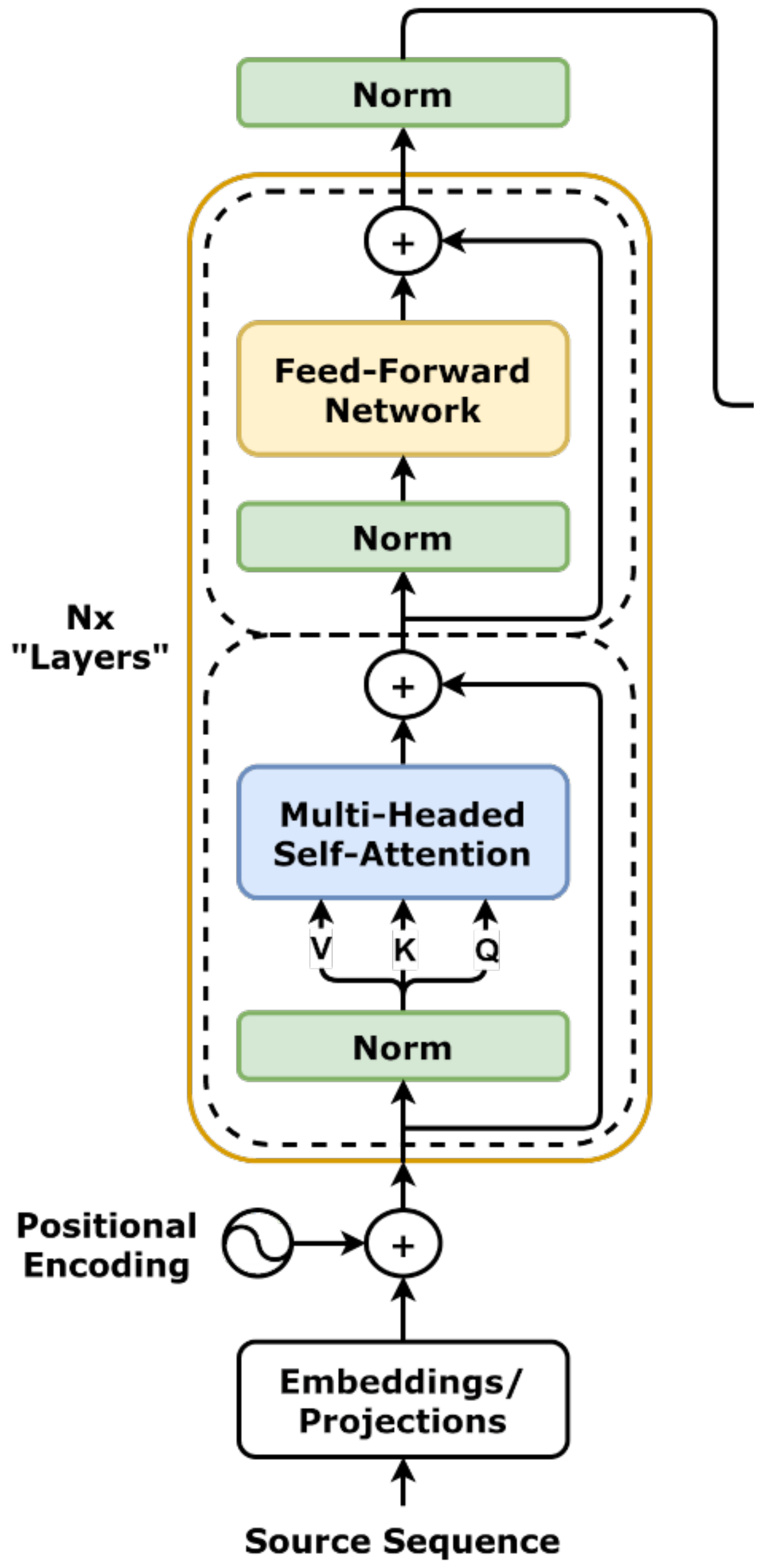
Attention Head

d_ff = hidden size of the feed-forward layer

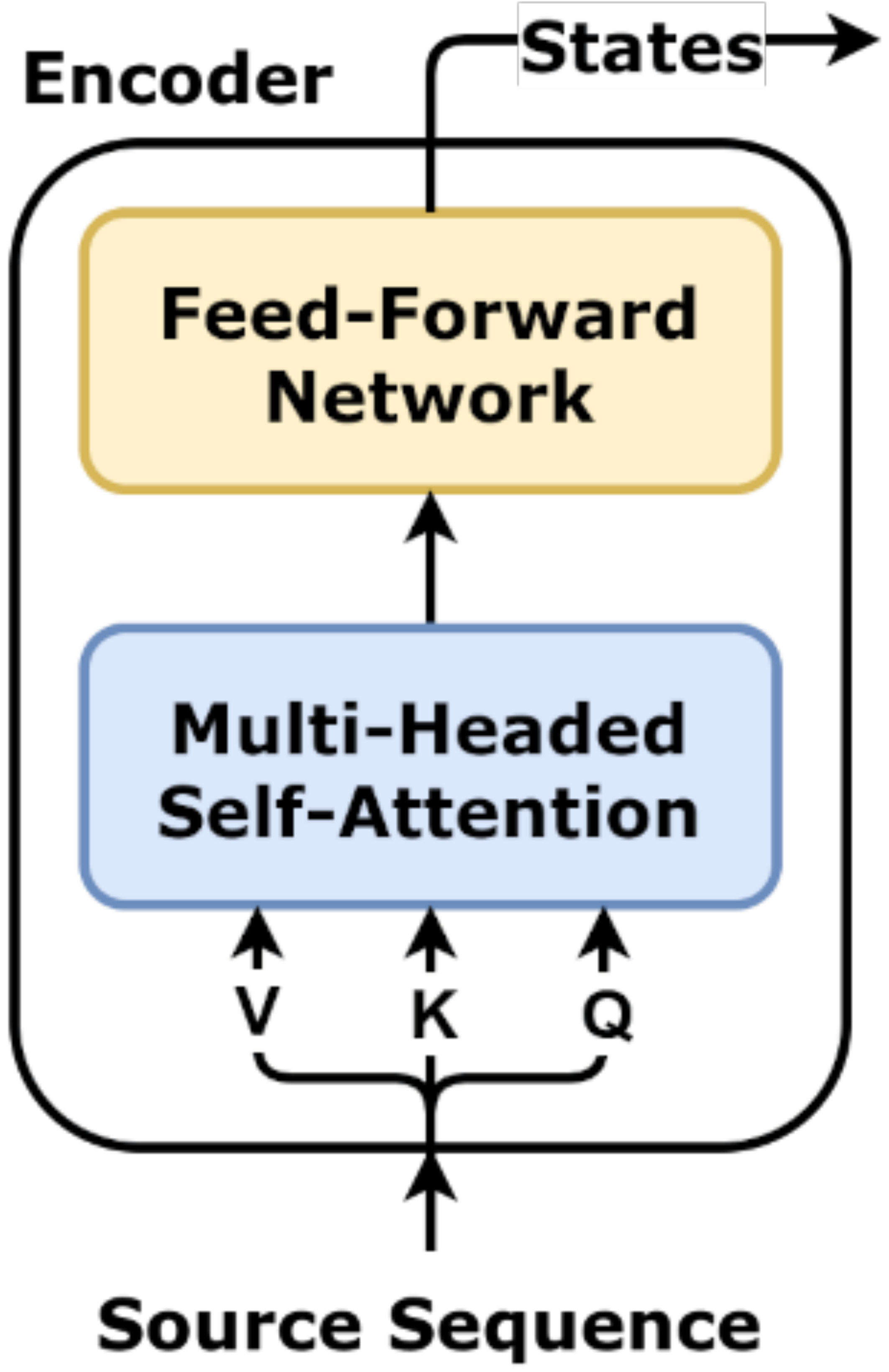
$$\mathbf{Z} = \mathrm{LayerNorm}\left[\mathbf{Y}(\mathbf{X}) + \mathbf{X}\right]$$

$$\widetilde{\mathbf{X}} = \mathrm{LayerNorm}\left[\mathrm{MLP}\left[\mathbf{Z}\right] + \mathbf{Z}\right]$$

MLP network with D_model inputs and D_model outputs

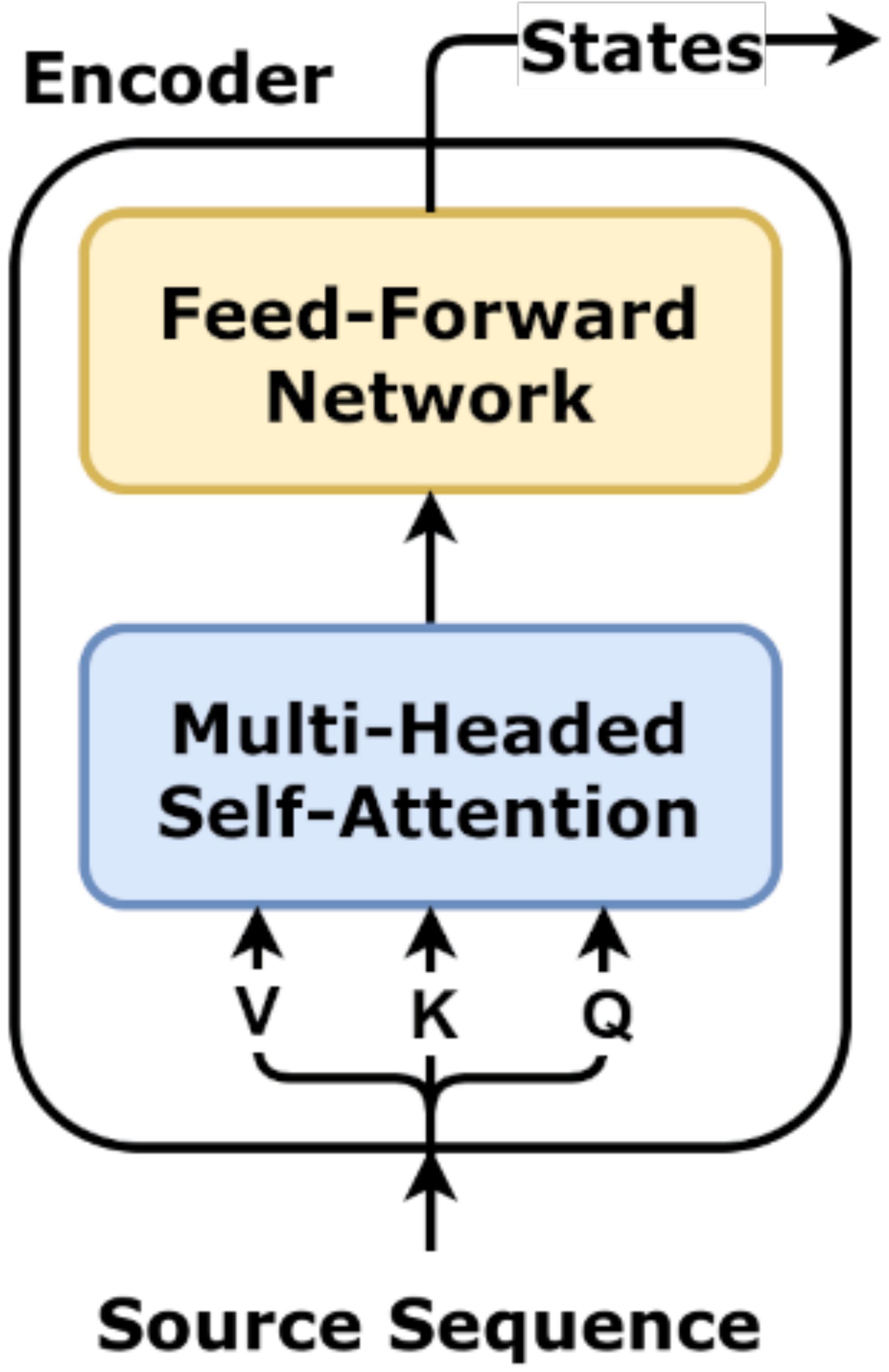


Training of an encoder-only model



Quiz : Implement training loss to learn a masked token in a sequence unsupervised ?

Training of an encoder-only model



Quiz : Implement training loss to learn a masked token in a sequence unsupervised ?

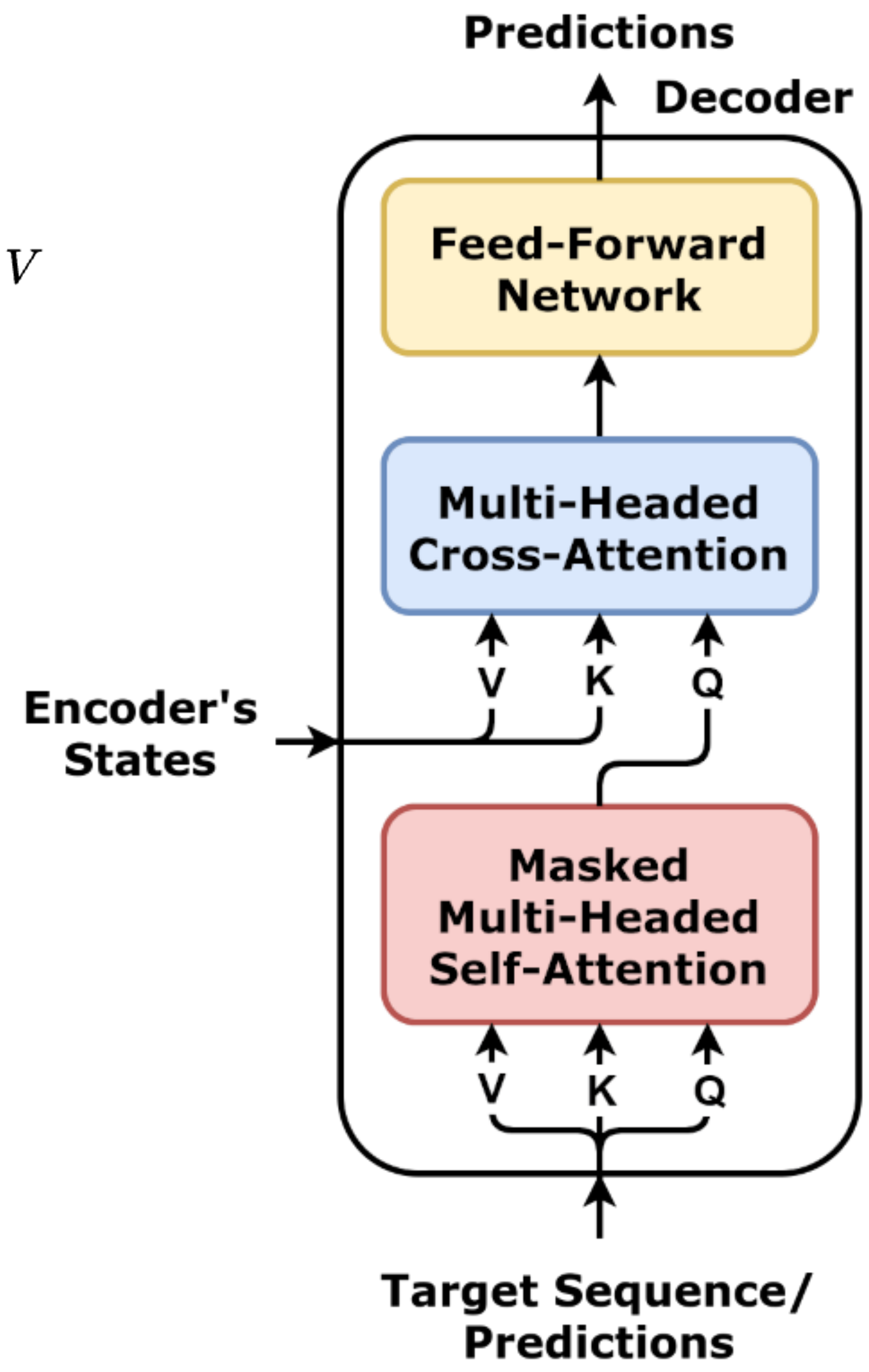
15% masked word : 80% masked, 10% wrong , 10% Unchanged

Logits = HW_dic

Decoder Transformer

$$\mathrm{MaskedAttention}(Q,K,V) = \mathrm{softmax}\left(M + \frac{QK^{\mathrm{T}}}{\sqrt{d_k}}\right)V$$

$$M_{\text{causal}} = \begin{bmatrix} 0 & -\infty & -\infty & \dots & -\infty \\ 0 & 0 & -\infty & \dots & -\infty \\ 0 & 0 & 0 & \dots & -\infty \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 0 \end{bmatrix}$$



Training strategy

Next Token prediction

SeqtoSeq : Average over all loss

Permutation Equivariant

If you permute the input tokens, the transformer produces outputs in the same permuted order

$$\text{Transformer}(PX) = P(\text{Transformer}(X))$$

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It is not a good feature for sequential data

No sense of order

Positional Encoding

X = emb + pos

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It is not a bad feature in Physics necessarily